

ONE-BUBBLE NEIGHBORHOODS OF SEIBERG–WITTEN STRATA IN $SO(3)$ -MONOPOLE MODULI SPACES

BERND JOHANNES WUEBBEN

ABSTRACT. Feehan and Leness constructed local gluing and obstruction maps near Seiberg–Witten strata in the Uhlenbeck compactification of the $SO(3)$ -monopole moduli space. Their level-one link calculation also requires continuity, embedding, neighborhood-surjectivity, and orientation properties assigned to a companion gluing theorem and retained as a hypothesis. We prove these properties at the one-bubble level over any compact regular Seiberg–Witten moduli space consisting of solutions with nonzero spinor.

A global equivariant stabilization of the normal deformation operators handles the family problem without a fixed spectral cutoff. A scale-uniform augmented parametrix in mixed critical norms gives $O(\lambda^{1/3})$ bounds for the projected splicing error and nonlinear correction, uniformly over the family. Model-coordinate modulation recovers the background and gluing parameters; a two-sided spectral estimate on the conformal neck gives the no-neck theorem needed for reverse gluing. The obstruction section extends over the natural cone family, and equivariant determinant-line excision identifies the actual and virtual links as oriented cycles.

Consequently, this proves the neighborhood-theorem input to the Feehan–Leness level-one pairing over regular strata of arbitrary dimension. We give its Jacobi-polynomial formula; the zero-dimensional formula and the associated low-degree Donaldson–Seiberg–Witten comparison follow as corollaries. The result is confined to one bubble and does not treat collision strata, zero-spinor solutions, or the full Witten conjecture.

1. INTRODUCTION

1.1. The neighborhood problem. Let X be a smooth, closed, connected, oriented four-manifold, and consider the Uhlenbeck compactification of a moduli space of $SO(3)$ monopoles. Its lower levels contain products of Seiberg–Witten moduli spaces with symmetric products of X . The $SO(3)$ -monopole cobordism program compares the link of the top anti-self-dual stratum with the links of these reducible strata. A link calculation has two distinct analytic inputs. One must first construct solutions of a finite-dimensional extension of the monopole equations from splicing data. One must then prove that the resulting zero set is an actual neighborhood of the ideal stratum in the Uhlenbeck topology.

The first input was established in the local gluing theorem of Feehan and Leness [FL99]. That theorem constructs the gluing-data bundle, a smooth solution map for the extended equations, and a finite-rank obstruction section. It gives the inclusion of the obstruction zero set in the monopole moduli space. Section 1.5.1 of the same paper separates from this construction the continuity, embedding, and neighborhood-surjectivity properties and assigns them to a companion article. In the one-bubble case, these properties are stated in [FL02, Theorem 3.8] and used in the oriented link calculation there. The global version appears as Hypothesis 7.8.1 in [FL18].

Date: 30 August 2026.

2020 Mathematics Subject Classification. 57K41 (primary); 57R57, 58J05, 58J37, 53C07 (secondary).

Key words and phrases. $SO(3)$ monopoles, Uhlenbeck compactification, gluing, Seiberg–Witten moduli spaces, Donaldson invariants, Witten conjecture.

This paper proves the missing neighborhood statement for one charge-one bubble over a compact regular Seiberg–Witten moduli space of arbitrary dimension. At this level there are no collision strata and no overlaps among partitions of a symmetric product. The family problem cannot be reduced to the zero-dimensional case by choosing one spectral cutoff: spectral flow can change the rank of the corresponding small-eigenvalue spaces. We instead stabilize the normal family globally by a smooth finite-rank equivariant bundle. Local cutoff reductions enter only through a common enlargement, so neither the gluing chart nor its orientation depends on a constant-rank spectral window.

1.2. Results. Let $b_1(X) = 0$ and let $b_2^+(X) \geq 3$ be odd. Fix generic perturbation data, a spin^u structure $\mathfrak{t}'$, and a spin^c structure $\mathfrak{s}$ for which the level-one reducible stratum is

$$(1) \quad M_{\mathfrak{s}} \times X \subset \overline{\mathcal{M}}_{\mathfrak{t}'}$$

Assume that $M_{\mathfrak{s}}$ is a compact regular smooth manifold, every represented Seiberg–Witten pair has nonzero spinor, and

$$(2) \quad \dim M_{\mathfrak{s}} = d_{\mathfrak{s}} = 2d \geq 0.$$

The level-one construction has a compactified virtual gluing space

$$(3) \quad \overline{\mathcal{M}}_{\mathfrak{t}',\mathfrak{s}}^{\text{vir}}(\varepsilon, \delta),$$

an obstruction pseudo-bundle $\overline{\Upsilon}$, and a section χ . Their definitions are recalled in Section 2. The instanton summand of $\overline{\Upsilon}$ has complex rank one at positive scale and collapses at the zero-scale cone point.

Writing $N(\varepsilon)$ for the background virtual-normal disk, the compactified gluing space has the three strata

$$(4) \quad \mathcal{G}^+ \sqcup \underbrace{(N(\varepsilon) - M_{\mathfrak{s}}) \times X}_{\text{middle zero-scale stratum}} \sqcup \underbrace{M_{\mathfrak{s}} \times X}_{\text{lowest stratum}}.$$

At positive scale the obstruction section has background and instanton components $\chi = (\chi_s, \chi_i)$. At zero scale the instanton fiber and χ_i collapse, while χ_s restricts to the background obstruction section $b : N(\varepsilon) \rightarrow \Xi$. Thus b is transverse on the middle stratum and vanishes on the lowest stratum.

Theorem A (level-one neighborhood; = Theorem 2.1). *For sufficiently small $\varepsilon, \delta > 0$, there is an S^1 -equivariant map*

$$\gamma : \overline{\mathcal{M}}_{\mathfrak{t}',\mathfrak{s}}^{\text{vir}}(\varepsilon, \delta) \longrightarrow \overline{\mathcal{C}}_{\mathfrak{t}'}$$

with the following properties.

- (1) *The map is a topological embedding, is smooth on the positive-scale stratum, and is joined to the splicing map by an S^1 -equivariant isotopy through compactified embeddings. On the zero-scale face it is the canonical background-and-ideal-point map.*
- (2) *There is an Uhlenbeck-open neighborhood $\mathcal{U}$ of $M_{\mathfrak{s}} \times X$ for which the restriction*

$$\gamma : \chi^{-1}(0) \cap \gamma^{-1}(\mathcal{U}) \longrightarrow \mathcal{U} \cap \overline{\mathcal{M}}_{\mathfrak{t}'}^{*,0}$$

is a homeomorphism and is a diffeomorphism on the positive-scale stratum.

- (3) *The total obstruction section is transverse on the positive-scale stratum, and its background component is transverse on the middle zero-scale stratum. Its norm is continuous along the compactified section. No transversality is asserted on the lowest stratum.*

- (4) *For a homology orientation of X , determinant-line excision gives the orientation of the transverse zero locus. This orientation descends through the frame and residual-circle quotients and induces the standard boundary orientation used in the level-one link pairing.*

The last qualification in (3) is forced by the geometry. On the lowest stratum the surviving background obstruction section is identically zero. If its rank is positive, its restriction cannot be transverse. This is the statement used in the relative-Euler-class calculation of [FL02, Section 5]; it also agrees with the qualification following Theorem 3.8 there. Under the fine decomposition (4), the all-strata transversality formulation in [FL18, Hypothesis 7.8.1(2)] would also include the lowest stratum and is false there when the background obstruction has positive rank.

The analytic estimate behind Theorem A may be stated independently. Write q_λ for the standard one-bubble pregluing and $J_{s,\lambda}$ for the matched embedding of the background obstruction space. If $\mathfrak{S}$ is the monopole map and $\mathfrak{S}(q) = \iota_{s,q}b(q)$ on the background virtual neighborhood, then

$$(5) \quad \|\mathfrak{S}(q_\lambda) - J_{s,\lambda}b(q)\|_{L^{\sharp,2;2}} \leq C\lambda^{1/3}.$$

There is a uniformly bounded inverse for the augmented linearization on a model-coordinate complement of all gluing parameters. The contraction therefore gives a unique correction and obstruction coefficients satisfying

$$(6) \quad \|u_\lambda\|_{L_1^2} + \|\eta_{s,\lambda}\| + \|\eta_{i,\lambda}\| \leq C\lambda^{1/3}.$$

The neighborhood theorem supplies every invocation of the deferred gluing result in Sections 3–6 of [FL02]. We retain the notation of that paper for the pairing. Let $\mathfrak{t}'$ be a spin^u structure with $w_2(\mathfrak{t}')$ good in the sense of [FL02, Definition 2.3], and suppose

$$(7) \quad (c_1(\mathfrak{t}') - c_1(\mathfrak{s}))^2 = p_1(\mathfrak{t}') + 4.$$

Let $h \in H_2(X; \mathbb{R})$ and let $x \in H_0(X; \mathbb{Z})$ be the positive generator. For a monomial z in h and x , let $\overline{V}(z)$ denote the closure of a transverse geometric representative of the Donaldson class $\mu_p(z)$; let $\overline{W}$ represent the circle class μ_c ; and let $\overline{L}_{\mathfrak{t}',\mathfrak{s}}$ be the standard oriented link of $M_{\mathfrak{s}} \times X$ in $\overline{\mathcal{M}}_{\mathfrak{t}'}^{*,0}/S^1$. The symbol $\#$ below denotes the signed count of a transverse zero-dimensional intersection. Let $\mu_{\mathfrak{s}} \in H^2(M_{\mathfrak{s}}; \mathbb{Z})$ be the first Chern class of the Seiberg–Witten base-point fibration. With the orientation determined by the chosen homology orientation, set

$$(8) \quad \mathrm{SW}_X(\mathfrak{s}) = \left\langle \mu_{\mathfrak{s}}^d, [M_{\mathfrak{s}}] \right\rangle.$$

Also write

$$(9) \quad d_a(\mathfrak{t}') = -2p_1(\mathfrak{t}') - \frac{3}{2}(\chi + \sigma),$$

and set

$$(10) \quad \beta = c_1(\mathfrak{s}) - c_1(\mathfrak{t}'), \quad B_h = \langle \beta, h \rangle.$$

For the Jacobi polynomials we use the normalization

$$(11) \quad P_d^{A,B}(\zeta) = 2^{-d} \sum_{v=0}^d \binom{d+A}{v} \binom{d+B}{d-v} (\zeta-1)^{d-v} (\zeta+1)^v.$$

Theorem B (family level-one pairing; = Theorem 8.2). *Let δ, m, η be nonnegative integers satisfying*

$$0 \leq m \leq \lfloor \delta/2 \rfloor, \quad 2(\delta + \eta) = \dim(\mathcal{M}_{\mathcal{V}}/S^1) - 1.$$

Put $n = \delta - 2m$ and

$$\begin{aligned} A_J &= \eta - d + 1, & B_J &= \frac{1}{2}(2\delta - d_a(\mathbf{t}') - d_s) - \frac{1}{4}(\chi + \sigma), \\ J_d &= P_d^{A_J, B_J}(0), & J'_d &= P_d^{A_J-1, B_J+1}(0). \end{aligned}$$

With the standard level-one link orientation,

$$\begin{aligned} &\#(\overline{V}(h^{\delta-2m}x^m) \cap \overline{W}^\eta \cap \overline{L}_{\mathcal{V}, \mathfrak{s}}) \\ &= (-1)^{m+1+d} 2^{d-\delta} \left\langle \mu_{\mathfrak{s}}^d, [M_{\mathfrak{s}}] \right\rangle \\ &\quad \times [J_d(a_0 B_h^n + a_1 B_h^{n-2} Q_X(h, h)) + 2n J'_d B_h^{n-1} \langle c_1(\mathbf{t}'), h \rangle], \end{aligned}$$

where terms with negative powers of B_h are omitted and

$$a_0 = 3\beta^2 + c_1^2(X) + 2\beta \cdot c_1(\mathbf{t}') + 4n - 4m, \quad a_1 = 4 \binom{n}{2}.$$

This is the polynomial form of [FL02, Theorem 6.1]; it remains meaningful when $J_d = 0$. The new neighborhood theorem supplies the deferred analytic hypothesis of that calculation. Its universal-class pullbacks, equivariant Thom and Segre reduction, charge-one fiber integrals, and Jacobi identity remain published inputs. When $d = 0$, one has $J_0 = J'_0 = 1$ and the theorem reduces exactly to the zero-dimensional formula proved in the earlier version of this manuscript.

For completeness, write

$$c(X) = -\frac{1}{4}(7\chi + 11\sigma).$$

Combining Theorem B with the published top-level calculation, blow-up formulas, and superconformal-simple-type vanishing gives the following precise consequence.

Theorem C (degree consequence; = Corollary 8.4). *Suppose in addition that X is abundant, effective, and of Seiberg–Witten simple type, and choose the characteristic data as in [FL02, Theorem 1.1]. Then*

$$\mathcal{D}_X^w(h) \equiv 0 \equiv \text{SW}_X^w(h) \pmod{h^{c(X)-2}}$$

and

$$\mathcal{D}_X^w(h) \equiv 2^{2-c(X)} e^{Q_X(h, h)/2} \text{SW}_X^w(h) \pmod{h^{c(X)+2}}.$$

The corollary retains every hypothesis just listed. It does not prove the full Witten conjecture, remove effectiveness, or treat a level $\ell \geq 2$ contribution.

1.3. Method. There are four analytic points. First, the compact family of background normal operators is stabilized by a global complex equivariant bundle Ξ . The resulting augmented operator contains both Ξ and the complex one-dimensional cokernel of the charge-one Dirac operator. A logarithmic square partition gives a cutoff error $C(\log N)^{-1/2}$. The physical bubble geometry is compactified to a family of metrics on S^4 ; using the induced operator eliminates a principal-symbol comparison which is not bounded in the critical target. All remaining coefficient errors are $O_N(\lambda^{1/3})$. The inverse and its first two normalized parameter derivatives are uniformly bounded on transported graph domains, which gives a smooth correction over the family.

Second, the complement is not the physical L^2 -orthogonal complement of the bubble Jacobi fields. Their physical L^2 norms collapse with scale. Instead, a glued variation is localized, transported to the fixed background and S^4 models, and projected there. Synthesis followed by this retraction is $I + o(1)$. It gives a uniform projection onto the background normal, translation, dilation, and relative-frame modes and a uniformly coercive complement.

Third, parameter recovery requires a slice estimate uniform both in the family and in the bubbling scale. We prove a critical Green estimate for the coupled pair gauge Laplacian. Exterior-based gauge normalization retains the constant adjoint mode at bubble infinity as the relative frame; normalizing at a finite point on the bubble would erase precisely that gluing parameter.

Fourth, continuity and compactness do not imply neighborhood surjectivity. For a genuine monopole sequence converging to the ideal stratum, a charge-one rescaling and a fixed-annulus estimate reduce the neck to a first-order equation on a conformal cylinder. The nonzero spectrum of the odd-signature and Dirac tangential operators is separated uniformly from zero. A two-sided Duhamel estimate then gives a length-independent ℓ^1 bound, including the $r^{3/2}$ outer spinor-cutoff term. Exact modulation places the resulting difference in the complement used for the construction. The augmented inverse and contraction uniqueness identify the original monopole with the gluing solution.

The determinant calculation follows the same decomposition. Excision splits the background and bubble determinant lines. The complex background normal and obstruction factors cancel in the finite-dimensional reduction, as do the tangent- and adjoint-frame quotient factors. The residual-circle and outward-radial conventions give

$$(12) \quad \lambda_{\mathcal{M}} = -v_{S^1} \wedge \vec{r} \wedge \tilde{\lambda}_L,$$

the convention in [FL02, Equation (3.81)].

1.4. Relation to prior work. The local construction quoted here is the one-bubble specialization of [FL99]. The critical norms and uniform slice estimates are from [Fee01]; the Uhlenbeck gauge theorem and compactness principles are used in their standard form [Uhl82, DK90]. The induced-metric construction follows the conformal comparison in [FL99, Section 8]. The parameter-recovery and reverse-gluing arguments are adapted to the coupled monopole equation rather than imported from ASD gluing: the bubble spinor, mixed critical target, and background obstruction must be controlled simultaneously.

Theorem A concerns the first lower Uhlenbeck level only. The topological splicing and overlap problem at all levels is treated in [FL18], while the analytic neighborhood identification is assumed there. At level one the overlap problem is absent, which is what makes it possible to separate and prove the neighborhood theorem without solving the collision problem. The varying-rank issue is handled here by global family stabilization and stable comparison with the local spectral reductions, not by assuming that a spectral window has constant rank.

2. LEVEL-ONE GLUING DATA

2.1. The monopole map and its deformation operator. Let $\mathfrak{t}'$ be represented by a Clifford module $V = V^+ \oplus V^-$ with Clifford multiplication ρ and adjoint bundle $\mathfrak{g}_V$. A configuration is a pair (A, Φ) consisting of a compatible connection and a section of V^+ . We use the perturbed monopole map

$$(13) \quad \mathfrak{S}(A, \Phi) = (F_A^+ - \tau \rho^{-1}(\Phi \otimes \Phi^*)_{00}, (D_A + \rho(\vartheta))\Phi).$$

The perturbations are fixed throughout and belong to a compact generic family. For the determinant-one gauge group, our infinitesimal-action convention is

$$d_{A,\Phi}^0 \zeta = (d_A \zeta, -\zeta \Phi).$$

Thus

$$d_{A,\Phi}^{0,*}(a, \phi) = d_A^* a - (\cdot \Phi)^* \phi,$$

and the derivative of (13) is

$$(14) \quad \begin{aligned} d_{A,\Phi}^1(a, \phi) &= (d_A^+ a - \tau \rho^{-1}(\Phi \otimes \phi^* + \phi \otimes \Phi^*)_{00}, \\ &\quad (D_A + \rho(\vartheta))\phi + \rho(a)\Phi). \end{aligned}$$

The gauge-fixed operator is

$$(15) \quad \mathcal{D}_{A,\Phi} = d_{A,\Phi}^{0,*} \oplus d_{A,\Phi}^1.$$

At an approximate solution $d^1 d^0$ need not vanish. All parametrices below include the gauge row and do not use a deformation complex at the preglued pair.

Equation (2.3) of [FL99] prints a factor 1/2 in the spinor-bilinear term of d^1 , whereas the literal derivative of the displayed monopole map, Equation (9.3) there, and [FL98, Equation (2.36)] have the normalization in (14). We use (14). Interpolating the discrepant zeroth-order coefficient through $[1/2, 1]$ does not change a determinant-line orientation; the characteristic-class normalization is checked independently in Section 8.

2.2. The background finite-dimensional reduction. The reducible splitting is

$$\mathfrak{t} = \mathfrak{s} \oplus \mathfrak{s} \otimes L.$$

Along the reducible embedding of $M_{\mathfrak{s}}$, the gauge-fixed family splits into tangential and normal summands

$$(16) \quad \mathcal{D} = \mathcal{D}^t \oplus \mathcal{D}^n.$$

Regularity of the Seiberg–Witten equations gives

$$(17) \quad \ker \mathcal{D}^t = TM_{\mathfrak{s}}, \quad \text{coker } \mathcal{D}^t = 0.$$

The normal family is complex linear, and the residual circle acts on its domain and target through one complex character.

We use the extrinsic finite-dimensional reduction of [FL01a, Theorem 3.19]. Compactness of $M_{\mathfrak{s}}$ gives a finite-rank complex bundle Ξ and an equivariant target embedding ι_{Ξ} such that

$$(18) \quad \widehat{\mathcal{D}}_p^n = \mathcal{D}_p^n \oplus (-\iota_{\Xi,p}) : \text{Dom } \mathcal{D}_p^n \oplus \Xi_p \longrightarrow \text{Tar } \mathcal{D}_p^n$$

is surjective for every $p \in M_{\mathfrak{s}}$. Indeed, each p admits a finite-dimensional complex cokernel-spanning space. Surjectivity persists on a neighborhood, a finite subcover supplies a common enlargement, and an equivariant tubular retraction extends it to a small normal neighborhood. No spectral projection is used in this construction.

The augmented kernels form a smooth complex bundle

$$(19) \quad N_{\mathfrak{t},\mathfrak{s}} = \ker \widehat{\mathcal{D}}^n \longrightarrow M_{\mathfrak{s}}.$$

Thus the full augmented kernel is $TM_{\mathfrak{s}} \oplus N_{\mathfrak{t},\mathfrak{s}}$ and

$$(20) \quad \text{Ind}_{\mathbb{C}}(\mathcal{D}^n) = [N_{\mathfrak{t},\mathfrak{s}}] - [\Xi] \quad \text{in } K^0(M_{\mathfrak{s}}).$$

The projected convention of [FL99] is equivalent: if $\Pi_{\Xi^{\text{perp}}}$ is orthogonal projection, then

$$(21) \quad u \longmapsto (u, \iota_{\Xi}^{-1} \mathcal{D}^n u)$$

identifies $\ker(\Pi_{\Xi^\perp} \mathcal{D}^n)$ with $\ker \widehat{\mathcal{D}}^n$.

After choosing a sufficiently small lifted disk in $N_{t,s}$, the implicit-function theorem gives a smooth embedding into the thickened configuration quotient and a section

$$(22) \quad b : N_{t,s}(\varepsilon) \longrightarrow \Xi, \quad \mathfrak{S}(q) = \iota_{s,q} b(q).$$

Its zero set is M_s . The term *virtual-normal disk* will always mean this disk bundle; it never means an eigenspace selected by a spectral cutoff. Since every pair on the zero section has nonzero spinor, compactness and openness allow the disk to be chosen so that every pair in it has trivial determinant-one infinitesimal stabilizer. Put

$$r_N = \text{rank}_{\mathbb{C}} N_{t,s}, \quad r_\Xi = \text{rank}_{\mathbb{C}} \Xi.$$

The background parameter bundle therefore has real dimension $d_s + 2r_N$, while its obstruction variables have real dimension $2r_\Xi$.

2.3. The charge-one factor and compactification. Let $M_1^{s,\natural}(S^4, \delta)$ be the centered framed charge-one ASD moduli space with scale in $(0, \delta]$. There are natural identifications

$$(23) \quad M_1^{s,\natural}(S^4, \delta) \cong (0, \delta] \times \text{SO}(3), \quad \overline{M}_1^{s,\natural}(S^4, \delta) \cong c(\text{SO}(3)).$$

If I is the standard centered instanton, then

$$(24) \quad \ker D_I = 0, \quad K_I := \text{coker } D_I = \ker D_I^*, \quad \dim_{\mathbb{C}} K_I = 1.$$

The framed ASD operator is surjective. Its eight-dimensional kernel consists of four translations, one dilation, and three framed rotations.

The pre-splicing factor is

$$(25) \quad \begin{aligned} \widetilde{\text{Gl}}_t(\delta) &= (\text{Fr}(\mathfrak{g}_V) \times_X \text{Fr}(T^*X) \times \text{Fr}(\mathfrak{g}_V|_s) \\ &\quad \times \widetilde{M}_1^\natural(S^4, \delta)) / (\text{SO}(3) \times \text{SO}(4)). \end{aligned}$$

After dividing by the charge-one gauge group, one obtains $\text{Gl}_t(\delta)$. Replacing the instanton factor by the cone in (23) gives $\overline{\text{Gl}}_t(\delta)$ and

$$(26) \quad \overline{\mathcal{M}}_{t',s}^{\text{vir}} = \widetilde{N}_{t,s}(\varepsilon) \times_{\mathcal{G}_s} \overline{\text{Gl}}_t(\delta).$$

At the cone point the relative instanton frame is forgotten. Hence (26) has three strata:

$$(27) \quad \begin{aligned} \overline{\mathcal{M}}_{t',s}^{\text{vir}} &= \mathcal{G}^+ \\ &\sqcup ((N_{t,s}(\varepsilon) - M_s) \times X) \sqcup (M_s \times X), \end{aligned}$$

where $\mathcal{G}^+$ denotes the positive-scale stratum.

Before quotienting, the splicing map is equivariant for the diagonal $\mathcal{G}_s$ and charge-one gauge actions and for the diagonal SO(4) tangent-frame and SO(3) adjoint-frame actions. These auxiliary frame actions are divided out; the relative SO(3) frame in (23) remains.

The residual circle has the two equivalent presentations of [FL02, Lemma 3.6]. It may act by the monopole circle on the normal factor and trivially on the compactified gluing factor, or diagonally with weight two on both. We use the effective circle in the link quotient. Passing from weight two to its effective parametrization does not reverse the orbit orientation.

2.4. The obstruction pseudo-bundle. At positive scale the obstruction space is

$$(28) \quad \Upsilon_g = \Upsilon_g^s \oplus \Upsilon_g^i \cong \Xi_q \oplus K_I.$$

The background summand extends as an ordinary bundle over (27). The instanton summand is the associated line

$$(0, \delta] \times \mathrm{SU}(2) \times_{\{\pm 1\}} \mathbb{C} \longrightarrow (0, \delta] \times \mathrm{SO}(3).$$

Its compactification is the cone map

$$(29) \quad c(\mathrm{SU}(2) \times_{\{\pm 1\}} \mathbb{C}) \longrightarrow c(\mathrm{SO}(3));$$

the fiber collapses at the vertex. This is a rank-dropping pseudo-bundle, not a singular trivialization of a line bundle.

2.5. Strong and weak levels. For smooth dependence at a fixed positive scale, choose $k \geq 4$ and use the standard Fredholm map

$$\mathcal{D}_{q_\lambda} : L_k^2(\Lambda^1 \otimes \mathfrak{g}_V \oplus V^+) \longrightarrow L_{k-1}^2(\mathfrak{g}_V \oplus \Lambda^+ \otimes \mathfrak{g}_V \oplus V^-).$$

Uniform estimates as $\lambda \downarrow 0$ use

$$(30) \quad \begin{aligned} \mathbb{X}_{q_\lambda} &= L_{1, A_\lambda}^2, \\ \mathbb{Z}_{q_\lambda} &= L^2(\mathfrak{g}_V), \\ \mathbb{Y}_{q_\lambda} &= L^{\sharp, 2}(\Lambda^+ \otimes \mathfrak{g}_V) \oplus L^2(V^-). \end{aligned}$$

Here $L^{\sharp, 2} = L^\sharp \cap L^2$ with the critical norm of [Fee01]. We construct the inverse on smooth target data, prove the estimate in (30), and extend by density. We do not assert that the full first-order operator maps every arbitrary L_1^2 section into the mixed critical target.

Choose matched finite-dimensional embeddings

$$J_{s, \lambda} : \Xi_q \rightarrow \mathbb{Y}_{q_\lambda}, \quad J_{i, \lambda} : K_I \rightarrow \mathbb{Y}_{q_\lambda}.$$

The augmented linearization is

$$(31) \quad \widehat{\mathcal{D}}_\lambda(u, \xi_s, \xi_i) = (d_{q_\lambda}^{0, *}, d_{q_\lambda}^1 u - J_{s, \lambda} \xi_s - J_{i, \lambda} \xi_i).$$

The nonlinear augmented map is

$$(32) \quad \begin{aligned} \widehat{\mathcal{F}}_\lambda(u, \eta_s, \eta_i) &= (d_{q_\lambda}^{0, *}, \mathfrak{S}(q_\lambda + u) \\ &\quad - J_{s, \lambda}(b(q) + \eta_s) - J_{i, \lambda} \eta_i). \end{aligned}$$

The retained parameter directions have real dimension $d_s + 2r_N + 8$: the background tangent and normal variables, four translations, dilation, and three relative rotations. The obstruction rank is $2r_\Xi + 2$. Hence the expected dimension of the obstruction zero set is

$$(33) \quad d_s + (2r_N + 8) - (2r_\Xi + 2) = d_s + 2(r_N - r_\Xi) + 6.$$

Theorem 2.1 (level-one neighborhood). *Under the hypotheses preceding Theorem A, there is an equivariant compactified embedding*

$$\gamma : \overline{\mathcal{M}}_{\nu, s}^{\mathrm{vir}}(\varepsilon, \delta) \longrightarrow \overline{\mathcal{C}}_\nu$$

and a continuous compactified obstruction section $\chi = (\chi_s, \chi_i)$ such that the following hold: the positive-scale maps are smooth; γ is equivariantly isotopic through embeddings to splicing; $\chi^{-1}(0)$ maps homeomorphically onto an Uhlenbeck-open neighborhood of $M_s \times X$ and diffeomorphically on the positive-scale stratum; χ is transverse there and χ_s is transverse on the middle zero-scale stratum; and the determinant-line orientation induces the standard level-one boundary orientation.

The proof occupies Sections 3–7. The local pregluing construction and the one-sided inclusion $\gamma(\chi^{-1}(0)) \subset \overline{\mathcal{M}}_{\nu}^{*,0}$ are quoted from [FL99]. Everything needed to turn that inclusion into the neighborhood homeomorphism is proved below.

3. THE UNIFORM AUGMENTED PARAMETRIX

3.1. Critical norms and the square partition. For a bundle-valued section f on X , put

$$(34) \quad \begin{aligned} \|f\|_{L^\sharp} &= \sup_{z \in X} \int_X \text{dist}(z, y)^{-2} |f(y)| dV(y), \\ \|f\|_{L^{2\sharp}} &= \sup_{z \in X} \left(\int_X \text{dist}(z, y)^{-2} |f(y)|^2 dV(y) \right)^{1/2}, \\ \|f\|_{L^{\sharp,2}} &= \|f\|_{L^\sharp} + \|f\|_{L^2}. \end{aligned}$$

The critical multiplication and Sobolev estimates of [Fee01, Lemmas 4.1 and 4.3] give

$$(35) \quad L_1^2 \hookrightarrow L^{2\sharp}, \quad L^{2\sharp} \cdot L^{2\sharp} \longrightarrow L^\sharp,$$

uniformly for the compact family of background connections under consideration. In particular,

$$(36) \quad \|v\|_{L^4} + \|v\|_{L^{2\sharp}} \leq C \|v\|_{L_{1,A}^2}.$$

Fix $N > 4$. The logarithmic cutoff of [FL99, Lemma 5.4], pulled back by geodesic normal coordinates at x , may be chosen to satisfy

$$(37) \quad \begin{aligned} \beta &= 0 & \text{if } r \leq N^{-1}\sqrt{\lambda}, & \quad \beta = 1 & \text{if } r \geq \tfrac{1}{2}\sqrt{\lambda}, \end{aligned}$$

$$\|d\beta\|_{L^4} \leq C(\log N)^{-3/4}, \quad \|d\beta\|_{L^{2\sharp}} \leq C(\log N)^{-1/2}, \quad \|d\beta\|_{L^2} \leq C(\log N)^{-1}\sqrt{\lambda}.$$

Define

$$(38) \quad \beta_X = \sin(\pi\beta/2), \quad \beta_I = \cos(\pi\beta/2).$$

Then $\beta_X^2 + \beta_I^2 = 1$, and both derivatives obey the bounds in (37). These estimates are uniform in x : near x this follows from normal-coordinate comparison, and for a base point outside a fixed larger chart the last, scale-decaying bound in (37) controls the supremum in (34).

Lemma 3.1 (cutoff commutator). *For $j = X, I$ and smooth $u = (a, \phi)$,*

$$(39) \quad [\mathcal{D}_{A_\lambda, \Phi_\lambda}, \beta_j]u = (-\iota_{\nabla\beta_j} a, (d\beta_j \wedge a)^+, \rho(d\beta_j)\phi),$$

and

$$(40) \quad \|[\mathcal{D}_{A_\lambda, \Phi_\lambda}, \beta_j]u\|_{\mathbb{Z} \oplus \mathbb{Y}} \leq C(\log N)^{-1/2} \|u\|_{\mathbb{X}_{q_\lambda}}.$$

The estimate extends by density to the weak domain.

Proof. All zero-order terms commute with a scalar cutoff. The three principal parts give (39). The gauge row, the L^2 part of the curvature row, and the Dirac row are bounded using $L^4 \cdot L^4 \subset L^2$ and the first estimate in (37). For the critical part of the curvature row, (35) gives

$$\|(d\beta_j \wedge a)^+\|_{L^\sharp} \leq C \|d\beta_j\|_{L^{2\sharp}} \|a\|_{L^{2\sharp}} \leq C(\log N)^{-1/2} \|a\|_{L_{1,A_\lambda}^2}.$$

Adding the rows proves (40). Notice that no derivative of $d\beta_j$ occurs: the augmented operator is first order. $\square$

3.2. The induced bubble operator. Let I be the centered, framed charge-one instanton on S^4 , and let $K_I = \text{coker } D_I$, a complex line. The stabilized round operator is

$$(41) \quad \widehat{B}_0(a, \phi, \kappa) = (d_I^* a, d_I^+ a, D_I \phi - \iota_I \kappa).$$

It is inverted on the fixed $L_1^2(S^4)$ complement of translations, dilation, and framed rotations.

A direct comparison of the physical operator with (41) would compare their principal symbols in $L^\sharp$. Such a comparison is not bounded merely by an L_1^2 domain norm. We instead compactify the physical normal-coordinate metric and its Clifford structure to a smooth metric $g_{x,\lambda}$ on S^4 , equal to the transported physical geometry on the active bubble region and to the round geometry near the north pole. Let $\widehat{B}_{x,\lambda}$ denote the operator defined by this geometry and the full instanton I .

In normal coordinates z at x , uniformly over X ,

$$(42) \quad g_{ij}(z) = \delta_{ij} + O(|z|^2), \quad \partial g_{ij}(z) = O(|z|), \quad \partial^2 g_{ij}(z) = O(1).$$

Put $z = \lambda y$ and $\Omega(y) = 2/(1 + |y|^2)$. On the active region $|y| \leq \frac{1}{2}\lambda^{-1/2}$, set

$$\widetilde{g}_{x,\lambda} = \Omega^2 g_x(\lambda y).$$

In the inverted coordinate $w = y/|y|^2$, its boundary has radius $2\sqrt{\lambda}$. Interpolate there to the round metric $g_0 = \Omega^2 \delta$ with a cutoff depending on $|w|/\sqrt{\lambda}$. The tensor being cut off is $O(\lambda)$, while one cutoff derivative costs $O(\lambda^{-1/2})$. Hence

$$(43) \quad \|\widetilde{g}_{x,\lambda} - g_0\|_{C^0(g_0)} \leq C\lambda, \quad \|\nabla^{g_0}(\widetilde{g}_{x,\lambda} - g_0)\|_{C^0(g_0)} \leq C\sqrt{\lambda}.$$

The interpolation lies where $\beta_I = 0$.

Proposition 3.2 (induced bubble inverse). *After fixing the stabilizing map ι_I , the operators $\widehat{B}_{x,\lambda}$ have right inverses $P_{x,\lambda}^I$ on the fixed model-kernel complement such that*

$$(44) \quad \|\widehat{B}_{x,\lambda} - \widehat{B}_0\|_{\mathcal{L}(L_1^2 \oplus K_I, L^2)} \leq C\sqrt{\lambda},$$

$$(45) \quad \|P_{x,\lambda}^I y\|_{L_1^2 \oplus K_I} \leq C\|y\|_{L^2}.$$

If I_λ^{tr} is the truncated instanton used in the pregluing, its replacement of I changes the transported operator by at most $C_N\sqrt{\lambda}$ from the weak domain to the mixed target.

Proof. Transport the physical zero-spinor ASD and Dirac blocks separately to $(S^4, \widetilde{g}_{x,\lambda})$. For the spinor block the conformal law is

$$D_A^{h^{-2}g} = h^{5/2} D_A^g h^{-3/2},$$

the ASD block has its four-dimensional conformal transport, and the Coulomb row is transported with the gauge-fixed ASD operator. On $\text{supp } \beta_I$ this definition is an exact conjugation of the physical operator. Thus its principal symbol agrees there exactly. Estimate (43) controls the principal coefficients in L^2 and the spin-connection and gauge-row zero-order coefficients, proving (44). The Neumann formula

$$P_{x,\lambda}^I = P_0^I (1 + (\widehat{B}_{x,\lambda} - \widehat{B}_0) P_0^I)^{-1}$$

then proves (45). It applies to mixed target data because that norm contains the L^2 norm. On the active region the induced and physical principal symbols agree exactly, so no principal-symbol error is estimated in $L^\sharp$.

In the south-pole framing, $|I(y)| \leq C|y|^{-3}$ and $|\nabla I(y)| \leq C|y|^{-4}$. Truncation begins at $R \asymp \lambda^{-1/2}$. If $b_\lambda = I_\lambda^{\text{tr}} - I$, radial integration gives

$$(46) \quad \|b_\lambda\|_{L^4} + \|\nabla b_\lambda\|_{L^2} \leq C\lambda, \quad \|b_\lambda\|_{L^2} + \|b_\lambda\|_{L^{2\sharp}} \leq C\sqrt{\lambda}.$$

Changing a connection changes all three rows only by multiplication by b_λ . The ordinary rows follow from $L^4 \cdot L^4 \subset L^2$; the critical curvature row follows from (35). This proves the last assertion. $\square$

3.3. Localized coefficients and matched obstruction spaces. The remaining comparison terms are zero-order. We repeatedly use the following two consequences of (34). If $f = (v, \psi)$ is a target section supported in $B(x, r)$ and is pointwise bounded by M , then

$$(47) \quad \|v\|_{L^{\sharp, 2}} + \|\psi\|_{L^2} \leq CMr^2.$$

Indeed, the L^2 bounds follow from volume, while the integral of $\text{dist}(z, \cdot)^{-2}$ over a four-ball is $O(r^2)$, uniformly in z .

If instead f is a bounded coefficient supported in the same ball, then

$$(48) \quad \|f\|_{L^4} + \|f\|_{L^{2\sharp}} \leq Cr\|f\|_{L^\infty}, \quad \|fu\|_{L^2} + \|fu\|_{L^\sharp} \leq Cr\|f\|_{L^\infty}\|u\|_{L_1^2}.$$

The first inequality follows by volume comparison and by splitting the supremum in (34) according as its base point lies within $2r$ of x ; the second follows from (35) and $L^4 \cdot L^4 \subset L^2$.

The pregluing formulas have the exact support properties

$$(49) \quad \begin{aligned} \Phi_\lambda &= \Phi_0 & \text{outside } B(x, 8\lambda^{1/3}), & \quad \Phi_\lambda = 0 & \text{inside } B(x, 4\lambda^{1/3}), \\ A_\lambda &= A_0 & \text{outside } B(x, 4\sqrt{\lambda}). \end{aligned}$$

Set $s_\lambda = \Phi_\lambda - \Phi_0$. The compact background family and (48) give

$$(50) \quad \|s_\lambda\|_{L^4} + \|s_\lambda\|_{L^{2\sharp}} \leq C\lambda^{1/3}.$$

This controls the spinor coefficients in the gauge, curvature, and Dirac rows, using L^4 for the ordinary rows and $L^{2\sharp}$ for the critical curvature row.

In radial gauge, $|A_0 - \Gamma| \leq C|z|$. On $\text{supp } \beta_X$, where $|z| \geq N^{-1}\sqrt{\lambda}$, the south-pole instanton decay gives

$$|I_\lambda(z)| \leq C\lambda^2|z|^{-3} \leq CN^3\sqrt{\lambda}.$$

Thus $c_{x,\lambda} = \beta_X(A_\lambda - A_0)$ satisfies

$$(51) \quad \|c_{x,\lambda}\|_{L^\infty} \leq C_N\sqrt{\lambda}, \quad \|c_{x,\lambda}\|_{L^4} + \|c_{x,\lambda}\|_{L^{2\sharp}} \leq C_N\lambda.$$

It contributes $O_N(\sqrt{\lambda})$ to the three rows, including the Coulomb row. Finally, the fixed perturbing one-form in the bubble Dirac row is supported by β_I in a ball of radius $\frac{1}{2}\sqrt{\lambda}$; ordinary Hölder gives an $O(\sqrt{\lambda})$ multiplier bound. Consequently the total localized coefficient error is

$$(52) \quad \|E_{\text{spin}} + E_{\text{conn}}^X + E_{\text{gauge}}^X\| \leq C_N\lambda^{1/3}.$$

Choose the metrics on Ξ_q and K_I so that the limiting embeddings $\iota_{s,q}$ and ι_I are isometries. With the same cutoffs as in (38), define

$$(53) \quad J_{s,\lambda}\xi = \beta_X T_{X,\lambda}^Y \iota_{s,q}\xi, \quad J_{i,\lambda}\kappa = T_{x,\lambda}^-(\beta_I^\# \iota_I \kappa),$$

where $\beta_I^\#(y) = \beta_I(\lambda y)$. The transports include the framing, dilation, and the negative-spinor conformal weight.

Lemma 3.3 (matched obstruction embeddings). *For all sufficiently small scales, $J_{s,\lambda} \oplus J_{i,\lambda}$ is injective with a uniformly bounded inverse on its image. Its L^2 projection is uniformly bounded in the mixed target norm. In the patched parametrix the two finite-dimensional stabilization terms cancel exactly.*

Proof. The background section removed by $1 - \beta_X$ is smooth and supported in a ball of radius $O(\sqrt{\lambda})$, so (47) shows that its mixed-target norm tends to zero. On the bubble, the cutoff complement moves to the north pole of S^4 ; compactness of the unit sphere in the finite-dimensional space K_I makes the L^2 tail uniformly small. The off-diagonal inner products tend to zero because the two supports separate after rescaling. The Gram matrix therefore tends uniformly to the identity. This proves the first two assertions. The last one follows directly from using precisely the profiles in (53) both when the model target is localized and when its obstruction coefficient is spliced back. Thus there is no separate obstruction error in the parametrix. $\square$

3.4. A complement which does not degenerate. Let

$$\mathcal{K}_X(q) = \ker \widehat{\mathcal{D}}_q^X, \quad \mathcal{K}_I = \ker \widehat{B}_0.$$

The first is identified by the background finite-dimensional reduction with $T_q N(\varepsilon)$ and has real dimension $d_s + 2r_N$: along the zero section its tangential summand is $T_p M_s$ and its normal summand is N_p . The second is the eight-dimensional framed ASD kernel. The physical L^2 norm of a naturally dilated bubble one-form is λ times its model norm. We therefore do not use physical L^2 -orthogonality to define the complement.

The parameter metric is

$$(54) \quad g_{\text{par},\lambda} = g_{\overline{N}} + \lambda^{-2} g_X + (d\lambda/\lambda)^2 + g_{\text{SO}(3)}.$$

Thus the normalized directions are $\lambda \partial_{x^j}$, $\lambda \partial_\lambda$, and ∂_{θ^k} . They converge after rescaling to a fixed basis of $\mathcal{K}_I$.

Define the synthesis map

$$(55) \quad S_{\lambda,N}(k_X, k_I) = \left(\beta_X T_{X,\lambda}^+ u_X + T_{I,\lambda}^+ (\beta_I^\# u_I), \eta_X, 0 \right),$$

where $k_X = (u_X, \eta_X)$ and $k_I = (u_I, 0)$. Conversely, localize a physical domain element by β_X and β_I , transport it to the two fixed models, project there onto the configuration components of $\mathcal{K}_X(q)$ and $\mathcal{K}_I$, and lift back to the augmented kernels. Denote the resulting retraction by $R_{\lambda,N}$. It ignores the two freely varying obstruction coefficients.

Let P_q^X denote the smoothly varying inverse of the stabilized background operator on the complement of $\mathcal{K}_X(q)$. We use this inverse, the kernel projection, and $\iota_{s,q}$ at the actual point q , rather than transport them to finitely many reference points. All their strong and mixed graph norms are uniformly bounded on the compact disk bundle. The background model equations therefore cancel exactly before the neck error is estimated.

Proposition 3.4 (model-coordinate complement). *For fixed N and sufficiently small λ ,*

$$(56) \quad M_{\lambda,N} = R_{\lambda,N} \text{pr}_{\mathbb{X}} S_{\lambda,N} = I + o(1)$$

uniformly in all gluing data. Hence

$$(57) \quad \Pi_{\lambda,N} = S_{\lambda,N} M_{\lambda,N}^{-1} R_{\lambda,N} \text{pr}_{\mathbb{X}}$$

is a uniformly bounded projection onto the $d_s + 2r_N + 8$ parameter modes. Moreover,

$$(58) \quad \|\widehat{\mathcal{D}}_\lambda S_{\lambda,N}\| \leq C(\log N)^{-1/2} + C_N \lambda^{1/3}.$$

Proof. On a background kernel vector, the part removed by the cutoff is supported in a shrinking ball. The derivative term is bounded by

$$\|d\beta_X u_X\|_{L^2} \leq C \|d\beta_X\|_{L^2} \|u_X\|_{L^\infty} \leq C_N \sqrt{\lambda} \|k_X\|.$$

On a bubble kernel vector, the normalized cutoff complement escapes to the north pole. Compactness of the unit sphere in the fixed finite-dimensional space gives uniform decay of its L_1^2 and L^4 tails. The two off-diagonal entries tend to zero: a smooth background mode acquires a positive power of λ under bubble rescaling, while only the escaping tail of a bubble mode reaches the background projection. This proves (56); it also gives the uniform bound in (57).

Apply the global augmented operator to (55). The two model kernel equations remove the interior terms, Lemma 3.1 controls the cutoffs, Proposition 3.2 and (52) control the model comparisons, and Lemma 3.3 cancels the stabilization terms. Their sum is (58). $\square$

Set

$$(59) \quad \mathbb{H}_{\lambda,N}^{\text{cmp}} = \ker(R_{\lambda,N} \text{pr}_{\mathbb{X}}).$$

Literal normalized derivatives of the pregluing map differ by $o(1)$ in L_1^2 from the synthesized modes, so they define the same uniformly transverse complement after decreasing the scale bound.

3.5. The exact inverse. Let $P_{x,\lambda}^I$ be the inverse from Proposition 3.2. We now give the raw parametrix explicitly. For a smooth physical target y , define

$$(60) \quad L_{X,\lambda}^Y y = (T_{X,\lambda}^Y)^{-1} (\beta_X y), \quad L_{I,\lambda}^Y y = (T_{I,\lambda}^Y)^{-1} (\beta_I y),$$

where the second transport includes dilation and the conformal spinor weight. The corresponding domain synthesis maps are

$$(61) \quad \begin{aligned} L_{X,\lambda}^X(u, \xi) &= (\beta_X T_{X,\lambda}^+ u, \xi, 0), \\ L_{I,\lambda}^X(v, \kappa) &= (T_{I,\lambda}^+ (\beta_I^\# v), 0, \kappa). \end{aligned}$$

Thus

$$(62) \quad P_{\lambda,N}^{\text{raw}} = L_{X,\lambda}^X P_q^X L_{X,\lambda}^Y + L_{I,\lambda}^X P_{x,\lambda}^I L_{I,\lambda}^Y.$$

Multiplication by the scalar cutoffs is bounded on the ordinary L^2 target rows. On the curvature row it is bounded on both L^2 and $L^\sharp$; the latter follows immediately from the defining positive-kernel integral in (34). The transports in (60) are uniformly bounded because the background geometry ranges over a compact family and the bubble transport is the exact conformal transport used to define $\widehat{B}_{x,\lambda}$ on the support of β_I . Equations (60)–(62) therefore define a uniformly bounded map from smooth mixed-target data to the augmented weak domain, and then by density to the graph closure. The parametrix used below is $(1 - \Pi_{\lambda,N}) P_{\lambda,N}^{\text{raw}}$.

An arbitrary L_1^2 correction need not have curvature row in $L^\sharp$. We therefore define $\mathbb{X}_{q\lambda}^{\text{gr}}$ to be the completion of smooth corrections and auxiliary coefficients in the norm

$$(63) \quad \begin{aligned} \|(u, \eta_s, \eta_i)\|_{\text{gr}} &= \|u\|_{L_1^2} + \|\eta_s\| + \|\eta_i\| \\ &\quad + \|\widehat{\mathcal{D}}_\lambda(u, \eta_s, \eta_i)\|_{\mathbb{Z} \oplus \mathbb{Y}}. \end{aligned}$$

The augmented operator is bounded from this graph domain by definition. Elliptic regularity identifies every graph-domain solution of the homogeneous equation with a strong Sobolev solution.

Theorem 3.5 (uniform augmented inverse). *There is $C > 0$ such that for every sufficiently large N there are $\varepsilon, \delta_N > 0$ for which*

$$(64) \quad \left\| \widehat{\mathcal{D}}_\lambda (1 - \Pi_{\lambda,N}) P_{\lambda,N}^{\text{raw}} - I \right\| \leq C(\log N)^{-1/2} + C_N \lambda^{1/3}.$$

For $q \in \overline{N}(\varepsilon)$ and $0 < \lambda \leq \delta_N$, the restriction

$$(65) \quad \widehat{\mathcal{D}}_\lambda : \mathbb{H}_{\lambda,N}^{\text{cmp}} \cap \mathbb{X}_{q\lambda}^{\text{gr}} \longrightarrow \mathbb{Z}_{q\lambda} \oplus \mathbb{Y}_{q\lambda}$$

is an isomorphism on the strong Fredholm level. Its inverse $P_{\lambda,N}$ obeys

$$(66) \quad \|P_{\lambda,N} y\|_{\text{gr}} \leq C \|y\|_{\mathbb{Z} \oplus \mathbb{Y}}.$$

The inverse is smooth and equivariant on every positive-scale stratum.

Proof. Apply $\widehat{\mathcal{D}}_\lambda$ to the two summands in (62). Their principal terms are $\beta_X^2 y$ and $\beta_I^2 y$, which recombine to y by (38). Lemma 3.1 contributes $C(\log N)^{-1/2}$; Proposition 3.2, (52), and variation of the background operator at the actual point q contribute $C_N \lambda^{1/3}$; the matched obstruction error is zero. Projecting the output adds at most the same quantity by (58). This proves (64).

Choose N first so that the logarithmic term is less than $1/4$, and then choose δ_N so that the scale term is less than $1/4$. A Neumann series gives a right inverse with image in (59) and proves (66). The strong augmented operator has index $d_s + 2r_N + 8$. The complement has codimension $d_s + 2r_N + 8$, because (56) is invertible. The restricted operator has index zero; surjectivity therefore implies injectivity. Elliptic regularity identifies the weakly constructed inverse with this strong inverse. The estimate extends to the mixed-target graph closure. Every ingredient is smooth and equivariant at positive scale, and inversion in the open group of bounded invertible operators preserves these properties. $\square$

Lemma 3.6 (normalized derivatives of the inverse). *Trivialize the mixed targets by the background and splicing transports and the complements by $1 - \Pi_{\lambda,N}$. For $j = 0, 1, 2$, covariant differentiation with respect to the normalized parameter metric (54) gives*

$$(67) \quad \|\nabla^j P_{\lambda,N}\|_{\mathcal{L}(\mathbb{Z} \oplus \mathbb{Y}, \mathbb{X}_{q\lambda}^{\text{gr}})} \leq C_j.$$

The same statement holds for the strong inverse in strong Sobolev norms.

Proof. Write $Q_\lambda = (1 - \Pi_{\lambda,N}) P_{\lambda,N}^{\text{raw}}$ and $E_\lambda = \widehat{\mathcal{D}}_\lambda Q_\lambda - I$. One or two normalized derivatives preserve the parametrix estimate. Background derivatives range over a compact unit sphere bundle; $\lambda \partial_{x_j}$ and $\lambda \partial_\lambda$ conjugate under bubble dilation to fixed derivatives of the unit-scale model; frame derivatives act in fixed compact-group representations; and differentiated cutoff commutators obey the same logarithmic bound. The same estimates hold for synthesis, retraction, and $\Pi_{\lambda,N}$. Consequently

$$\|\nabla^j E_\lambda\| \leq C_j (\log N)^{-1/2} + C_{j,N} \lambda^{1/3}, \quad 0 \leq j \leq 2.$$

Since $P_{\lambda,N} = Q_\lambda (I + E_\lambda)^{-1}$, differentiate this identity and use

$$\nabla(I + E)^{-1} = -(I + E)^{-1} (\nabla E) (I + E)^{-1}.$$

The second derivative contains the corresponding $\nabla^2 E$ term and two ordered products of ∇E . This proves the graph-domain and strong bounds. $\square$

In particular, a normalized sequence in the complement cannot be an approximate kernel supported in the neck: applying (66) would give $1 \leq o(1)$. This also excludes gauge modes. On the background the nonzero spinor removes the determinant-one stabilizer, and on the bubble it is removed by the based framing convention.

4. PROJECTED RESIDUAL AND NONLINEAR CORRECTION

4.1. The seven-region estimate. Put $r = \sqrt{\lambda}$ and $R = \lambda^{1/3}$. For small λ the pregluing formula divides X into the core $B(x, r/4)$; the inner, flat, and middle annuli with radii $(r/4, r/2)$, $(r/2, 2r)$, and $(2r, 4r)$; the zero-spinor plateau $(4r, 4R)$; the outer spinor-cutoff annulus $(4R, 8R)$; and the exterior of $B(x, 8R)$. The logarithmic parametrix cutoffs lie inside the r -scale annuli but play no role in this residual decomposition.

Proposition 4.1 (projected residual). *Uniformly in the compact background family, the center, and the frames,*

$$(68) \quad \|\mathfrak{S}(q_\lambda) - J_{s,\lambda}b(q)\|_{L^{\sharp,2;2}} \leq C\lambda^{1/3}.$$

Here $L^{\sharp,2;2}$ means $L^{\sharp,2}$ in the curvature component and L^2 in the negative-spinor component.

Proof. On the exterior, the preglued pair equals the background pair and $J_{s,\lambda}$ equals the background obstruction embedding; the residual therefore vanishes identically.

Write $\mathfrak{S}(q) = (v_0, \psi_0)$ and let χ be the outer spinor cutoff. On $4R < d(x, \cdot) < 8R$ the pair is $(A_0, \chi\Phi_0)$ and $J_{s,\lambda}b(q) = \mathfrak{S}(q)$. If $\mathcal{Q}(\Phi)$ denotes the quadratic spinor term in the curvature equation, the residual is

$$(69) \quad (\mathcal{Q}(\Phi_0) - \mathcal{Q}(\chi\Phi_0), (\chi - 1)\psi_0 + \rho(d\chi)\Phi_0).$$

The first component and the first term in the second component are uniformly bounded on a ball of radius $8R$, and hence have mixed-target norm $O(R^2)$ by (47). Direct radial integration gives $\|d\chi\|_{L^2} \leq CR$; consequently the last term in (69) is $O(R) = O(\lambda^{1/3})$.

On the plateau $4r < d(x, \cdot) < 4R$ the pair is $(A_0, 0)$, not a solution of the background Dirac equation. Here the exact difference from the matched background residual is

$$(70) \quad \mathfrak{S}(A_0, 0) - \mathfrak{S}(A_0, \Phi_0) = (\mathcal{Q}(\Phi_0), -\psi_0),$$

with the sign in $\mathcal{Q}$ fixed by the convention in (69). Both entries are uniformly bounded and supported in $B(x, 4R)$, so their mixed-target norm is $O(R^2) = O(\lambda^{2/3})$.

On the middle annulus $2r < d(x, \cdot) < 4r$, write the connection in radial gauge as $\Gamma + \chi_m(A_0 - \Gamma)$. Then

$$(71) \quad F = \chi_m F_{A_0} + d\chi_m \wedge (A_0 - \Gamma) + (\chi_m^2 - \chi_m)(A_0 - \Gamma) \wedge (A_0 - \Gamma).$$

The bounds $|A_0 - \Gamma| \leq Cr$ and $|d\chi_m| \leq C/r$ show that (71) is uniformly bounded. The spinor is zero and the matched background target is uniformly bounded, so (47) gives an $O(r^2) = O(\lambda)$ contribution. On the flat annulus $r/2 < d(x, \cdot) < 2r$ the preglued pair is $(\Gamma, 0)$; the only term after subtraction is again a uniformly bounded matched background target, and its contribution is $O(\lambda)$.

On the inner annulus $r/4 < d(x, \cdot) < r/2$, the instanton decay gives

$$(72) \quad |F_I| \leq C\lambda^2 d(x, \cdot)^{-4} \leq C, \quad |A_I| \leq C\lambda^2 d(x, \cdot)^{-3} \leq C\sqrt{\lambda}.$$

Since the cutoff derivative is $O(\lambda^{-1/2})$, every curvature term in the truncated connection and every subtracted background term is uniformly bounded. This annulus therefore also contributes $O(\lambda)$.

Finally, on $B(x, r/4)$ the model instanton is anti-self-dual for the Euclidean metric δ , but not identically for the physical metric g . Its self-dual curvature is

$$(73) \quad F_I^{+,g} = \frac{1}{2}(*_g - *_\delta)F_I.$$

The normal-coordinate estimate $|g - \delta| \leq Cd(x, \cdot)$, conformal invariance of the four-dimensional curvature norm, and radial integration of the standard instanton give

$$(74) \quad \|F_I^{+,g}\|_{L^{\sharp,2}(B(x,r/4))} \leq C\sqrt{\lambda}.$$

The matched background target on the core contributes $O(r^2) = O(\lambda)$. Adding the seven disjoint regional estimates gives

$$C(\lambda^{1/3} + \lambda^{2/3} + \lambda + \lambda^{1/2}) \leq C\lambda^{1/3},$$

which proves (68). $\square$

4.2. Solving the augmented equation. Write $h = (u, \eta_s, \eta_i)$ and let P_λ be the inverse in Theorem 3.5. Expanding (32) gives

$$(75) \quad \widehat{\mathcal{F}}_\lambda(h) = r_\lambda + \widehat{\mathcal{D}}_\lambda h + Q_\lambda(u, u),$$

where r_λ is the residual in Proposition 4.1. Sobolev multiplication gives the scale-independent estimates

$$(76) \quad \begin{aligned} \|Q_\lambda(u, u)\|_{\mathbb{Z} \oplus \mathbb{Y}} &\leq C\|u\|_{L_1^2}^2, \\ \|Q_\lambda(u, u) - Q_\lambda(v, v)\|_{\mathbb{Z} \oplus \mathbb{Y}} &\leq C(\|u\|_{L_1^2} + \|v\|_{L_1^2})\|u - v\|_{L_1^2}. \end{aligned}$$

Proposition 4.2 (small correction). *For sufficiently small λ , there is a unique $h_\lambda = (u_\lambda, \eta_{s,\lambda}, \eta_{i,\lambda})$ in a fixed small ball in $\mathbb{H}_{\lambda,N}^{\text{cmp}}$ satisfying $\widehat{\mathcal{F}}_\lambda(h_\lambda) = 0$. It obeys*

$$(77) \quad \|u_\lambda\|_{L_1^2} + \|\eta_{s,\lambda}\| + \|\eta_{i,\lambda}\| \leq C\lambda^{1/3}.$$

It is smooth in all positive-scale gluing parameters.

Proof. On the ball of radius $2C\|r_\lambda\|$, use the map

$$h \longmapsto -P_\lambda(r_\lambda + Q_\lambda(u, u)).$$

Theorem 3.5, Proposition 4.1, and (76) show that it preserves this ball and has Lipschitz constant less than 1/2 after decreasing the scale. Banach's fixed-point theorem gives existence, uniqueness, and (77). The strong Sobolev inverse and the implicit-function theorem give smooth dependence for $\lambda > 0$. $\square$

Define the corrected gluing map and obstruction section by

$$(78) \quad \begin{aligned} \gamma(g) &= [q_\lambda + u_\lambda], \\ \chi(g) &= (b(q) + \eta_{s,\lambda}, \eta_{i,\lambda}). \end{aligned}$$

Then $\gamma(g)$ is a genuine monopole exactly when $\chi(g) = 0$.

4.3. Continuity at zero scale. The constants above are independent of a moving center. If $g_n \rightarrow g_0$ with $\lambda_n \rightarrow 0$, then (77) gives $u_{\lambda_n} \rightarrow 0$ in the global weak norm. On compact subsets away from the limiting center, the preglued pairs converge in every Sobolev norm to the background. After rescaling at the centers, their connections converge to the framed instanton while their spinors converge to zero. These two statements, together with the energy identity, are precisely Uhlenbeck convergence to (q, x) . Thus γ extends continuously by

$$(79) \quad \gamma(q, x, 0) = (q, x).$$

The same estimates and the definitions in (53) show that the norm of χ is continuous when the instanton line is collapsed at the cone vertex.

5. PARAMETER RECOVERY AND THE COMPACTIFIED EMBEDDING

5.1. Uniform modulation. Work first on the framed cover of the positive-scale gluing domain. At a preglued pair $q_\lambda(g)$ impose the based Coulomb condition

$$(80) \quad d_{q_\lambda(g)}^{0,*}((A, \Phi) - q_\lambda(g)) = 0.$$

For $q_g = (A_g, \Phi_g)$, put

$$(81) \quad \Delta_g^0 = d_g^{0,*} d_g^0 = d_{A_g}^* d_{A_g} + \mathfrak{r}_{\Phi_g}^* \mathfrak{r}_{\Phi_g}.$$

Let $L_{2,g}^{\sharp,2}$ be the completion of smooth determinant-one infinitesimal gauges in the norm

$$(82) \quad \|\zeta\|_{2,\sharp,g} = \|\Delta_g^0 \zeta\|_{L^{\sharp,2}} + \|\zeta\|_{L^2}.$$

Feehan's critical embedding gives $L_{2,g}^{\sharp,2} \hookrightarrow C^0 \cap L_2^2$ [Fee01, Lemmas 5.3–5.6].

Theorem 5.1 (uniform critical slice for pairs). *After decreasing the maximum scale, there is $C > 0$ such that*

$$(83) \quad \|\zeta\|_{2,\sharp,g} \leq C \|\Delta_g^0 \zeta\|_{L^{\sharp,2}}$$

for every positive-scale family datum. The Green operator $G_g = (\Delta_g^0)^{-1}$ and its first two normalized parameter derivatives are uniformly bounded. Moreover, there is $r_{\text{sl}} > 0$, independent of the datum, such that every nearby based gauge orbit has a unique representative satisfying (80).

Proof. We first prove a uniform ordinary spectral gap. Otherwise there are data g_j and infinitesimal gauges ζ_j with $\|\zeta_j\|_{L^2} = 1$ and

$$\|d_{A_{g_j}} \zeta_j\|_{L^2} + \|\zeta_j \Phi_{g_j}\|_{L^2} \rightarrow 0.$$

Kato's inequality and Sobolev embedding give a common L^4 bound. Away from the bubbling point, Rellich compactness gives a limit that is parallel and annihilates the nonzero limiting spinor; hence it vanishes. No L^2 mass can remain at the point, since

$$\|\zeta_j\|_{L^2(B(x_j, 2r))} \leq Cr \|\zeta_j\|_{L^4(B(x_j, 2r))} \leq Cr.$$

This contradicts the normalization and proves the gap.

For $f = \Delta_g^0 \zeta$, positivity of $\mathfrak{r}_{\Phi_g}^* \mathfrak{r}_{\Phi_g}$ and the weak Kato inequality give

$$(d^* d + 1)|\zeta| \leq |f| + |\zeta|$$

in distributions. The scalar Green-kernel estimate and the spectral gap therefore yield

$$\|\zeta\|_{C^0} + \|\zeta\|_{L^2} \leq C \|f\|_{L^{\sharp,2}}.$$

Applying the covariant-Laplacian estimates of [Fee01, Lemmas 5.5–5.6] to $d_{A_g}^* d_{A_g} \zeta = f - \mathfrak{r}_{\Phi_g}^* \mathfrak{r}_{\Phi_g} \zeta$ proves (83). Direct differentiation of (81), followed by the critical multiplication estimates and $\nabla G = -G(\nabla \Delta^0)G$, proves the first two normalized derivative bounds. The normalization in (54) cancels the inverse lengths of the center and scale profiles.

The contraction argument for the gauge equation gives existence of a Coulomb representative with a uniform radius. For injectivity, suppose $q_g + c_1$ and $q_g + c_2$ are in the slice and related by a based gauge u ; set $v = u - 1$. The transition equations and the two Coulomb equations give the exact identity

$$(84) \quad \Delta_g^0 v = d_{A_g}^* (v a_1 - a_2 v) - \mathfrak{r}_{\Phi_g}^* (v \phi_1).$$

Pairing with v , using the ordinary gap just proved and Hölder's inequality, gives

$$\|d_{A_g} v\|_{L^2}^2 + \|v \Phi_g\|_{L^2}^2 \leq C(\|c_1\|_{L^2_1} + \|c_2\|_{L^2_1})(\|d_{A_g} v\|_{L^2}^2 + \|v \Phi_g\|_{L^2}^2).$$

Decrease r_{sl} and absorb. Then $v = 0$ by the gap, and the two representatives agree. This also excludes a slowly rotating transition along the long neck. Smooth dependence follows from the same contraction; weak continuity follows from the uniform bound and uniqueness. $\square$

The Coulomb-horizontal projection is

$$(85) \quad H_g = 1 - d_g^0 G_g d_g^{0,*}.$$

It and its first two normalized derivatives are uniformly bounded by Theorem 5.1.

Put

$$(86) \quad R_g = R_{\lambda,N} \text{pr}_{\mathbb{X}}, \quad \mathcal{C}_g^{\text{sl}} = \ker R_g \cap \ker d_{q_\lambda(g)}^{0,*}.$$

After locally trivializing this bundle, define the quotient-valued tubular map

$$(87) \quad \mathcal{T}(g, h) = [q_\lambda(g) + h], \quad h \in \mathcal{C}_g^{\text{sl}}.$$

Its differential along the zero section is

$$(88) \quad D\mathcal{T}_{(g,0)}(\dot{g}, \dot{h}) = H_g d q_\lambda(\dot{g}) + \dot{h}.$$

Let $j_g : T_g \mathcal{G}^+ \rightarrow \mathcal{K}_X(q) \oplus \mathcal{K}_I$ be the coordinate identification which sends the normalized parameter basis from (54) to the background virtual-normal modes and the translation, dilation, and framed-rotation Jacobi fields. Differentiating the splicing formulas gives

$$(89) \quad \|R_g H_g d q_\lambda - j_g\| \longrightarrow 0 \quad \text{uniformly as } \lambda \downarrow 0.$$

For background variations, the discrepancy in (89) is supported on a shrinking ball. For $\lambda \partial_x$, $\lambda \partial_\lambda$, and ∂_θ , rescaling gives the fixed instanton Jacobi fields. Derivatives of an $r = \sqrt{\lambda}$ cutoff contribute the compensating factors $\lambda/r = \sqrt{\lambda}$ and $\lambda^2/r^2 = \lambda$ for one and two normalized center derivatives; logarithmic-scale derivatives preserve the fixed cutoff profiles. The remaining terms are smooth background terms on shrinking annuli or escaping instanton tails. The estimates used in Proposition 3.4 apply to each such derivative. Horizontal projection changes the result by $o(1)$ because the gauge row has the uniform coercive bound just established. This proves (89).

Since $R_g \dot{h} = 0$, applying R_g to (88) first recovers $\dot{g}$ with a uniformly bounded inverse; substitution then recovers $\dot{h}$. Thus $D\mathcal{T}_{(g,0)}$ is uniformly invertible. The preceding differentiated cutoff estimates also give a uniform bound for the normalized second derivative of $\mathcal{T}$. The quantitative Banach inverse-function theorem, in the fixed local trivializations, consequently gives constants $c_{\text{mod}}, r_{\text{mod}} > 0$ independent of sufficiently small scale.

Lemma 5.2 (uniform modulation). *The map (87) is one-to-one on*

$$(90) \quad d_{\text{par}}(g, g') < c_{\text{mod}}, \quad \|h\|_{L_1^2} + \|h'\|_{L_1^2} < r_{\text{mod}}.$$

Equivalently, after the unique based gauge in (80) is imposed, every class in this tube has a unique parameter g characterized by

$$(91) \quad R_g((A, \Phi) - q_\lambda(g)) = 0.$$

The recovered parameter and based gauge are continuous in the weak topology and smooth on the positive-scale strong Sobolev charts. The construction is equivariant for the diagonal frame actions and therefore descends to the gluing quotient.

The correction in Proposition 4.2 satisfies both equations in (86). Hence exact modulation recovers the input parameter of every corrected configuration: no post-correction shift of center, scale, or relative frame is hidden in the construction.

5.2. Center, scale, and frame as observables. For an exact framed finite-energy connection on $\mathbb{R}^4$, the model center and scale are

$$(92) \quad z[A] = \frac{1}{8\pi^2} \int_{\mathbb{R}^4} y |F_A|^2, \quad \sigma[A]^2 = \frac{1}{8\pi^2} \int_{\mathbb{R}^4} |y - z[A]|^2 |F_A|^2.$$

They identify translations and dilations of the charge-one instanton. For the corrected configurations we use only bounded local observables: a fixed fractional-energy radius, a first moment truncated in normalized coordinates, and convergence of the rescaled framed connection on compact sets. Proposition 4.2 implies that these observables tend to those of the input instanton. We do not apply the unbounded global moments in (92) directly to the corrected connection.

Lemma 5.3 (outer-normalized bubble framing). *Let $q_{g_n} + u_n$ be corrected configurations with $\lambda_n \downarrow 0$ and*

$$\|u_n\|_{L_1^2} + \|\eta_{s,n}\| + \|\eta_{i,n}\| \leq C\lambda_n^{1/3}.$$

Normalize the exterior gauges by the fixed based background slice. The rescaled gauges may then be chosen to converge on compact subsets of $S^4 - \{\infty\}$. In temporal gauge the constant cylinder mode determines a value at bubble infinity, modulo the center, and this value is precisely the relative frame. The resulting outer-normalized framed bubble limit is unique.

Proof. Use the dyadic Coulomb gauges and the stabilized long-neck estimate of Proposition 6.4 below. Starting with the identity at the exterior end, remove the nonconstant temporal-gauge modes while retaining the constant endpoint $\rho_n \in \text{SO}(3)$. If k_j relates the remaining gauges on successive slabs, the fixed-annulus estimate gives

$$\text{dist}(k_{j+1}, k_j) \leq C(\alpha_j + \alpha_{j+1} + \beta_j + \beta_{j+1}),$$

where α_j and β_j are the connection and spinor norms on that slab. The two-sided cylinder estimate gives a length-independent bound for the sum of these quantities, tending to zero after the exterior and interior ends are sent out. Thus every nonconstant change of gauge on the neck vanishes in the limit, while compactness of $\text{SO}(3)$ gives a limit of the retained endpoints. Fixed-annulus elliptic estimates and the transition equation $dv_n = v_n a_n - a_n' v_n$ give convergence of the relating gauges on compact sets. Two outer-normalized limits differ by a stabilizer of the irreducible instanton with the same value at infinity, hence only by the central action already divided out. $\square$

Proposition 5.4 (separation of parameters). *Suppose g_n and g'_n are gluing parameters tending to the compactified domain and $\gamma(g_n) = \gamma(g'_n)$. Then their background points and ideal centers agree in the limit. If the scales are positive, their scale ratios tend to one and their relative frames tend to the same element of $\mathrm{SO}(3)$. For all sufficiently small scale, $g_n = g'_n$.*

Proof. If the scales stay bounded away from zero, the smooth local inverse in Lemma 5.2 gives the result. It remains to consider $\lambda_n, \lambda'_n \rightarrow 0$. Uhlenbeck convergence first identifies the two background limits and ideal points. The fixed background slices then give $d_N(q_n, q'_n) \rightarrow 0$.

Rescale the common corrected configuration about x_n by λ_n . The first representation converges on compact sets to the centered unit-scale instanton I . In these coordinates, the second bubble has relative translation and scale

$$(93) \quad z_n = \lambda_n^{-1} \exp_{x_n}^{-1}(x'_n), \quad t_n = \lambda'_n / \lambda_n.$$

If $|z_n| \rightarrow \infty$, the second description has no bubble energy on a fixed compact set. If $t_n \rightarrow \infty$, it becomes flat there; if $t_n \rightarrow 0$, its curvature measure tends to a point mass. Each alternative contradicts the smooth unit-scale limit. Thus, after a subsequence, $z_n \rightarrow z \in \mathbb{R}^4$ and $t_n \rightarrow t \in (0, \infty)$.

The two rescaled limits are gauge equivalent, and the second is the translate and dilate of I by (z, t) . Applying (92) to these exact limiting instantons gives $z = 0$ and $t = 1$. Apply Lemma 5.3 to both descriptions, using the same exterior based normalization. Their retained values at bubble infinity coincide, so their relative $\mathrm{SO}(3)$ frames agree modulo exactly the central and diagonal actions already divided out. Consequently

$$(94) \quad d_{\mathrm{par}}(g_n, g'_n) \rightarrow 0.$$

For large n , the two parameters lie within the uniform modulation radius. Put the common configuration in the based slice and apply Lemma 5.2; uniqueness gives literal equality. At zero scale the cone quotient has already identified all relative frames. $\square$

Because the compactified parameter space over the compact background is compact after the outer boundary is fixed, sequential separation also rules out coincidences between parameters that are not initially close. We have therefore proved the following statement.

Corollary 5.5 (compactified embedding). *After decreasing ε and δ , γ is an equivariant topological embedding of the compactified virtual gluing space. On the positive-scale stratum it is a smooth embedding.*

Finally write $\gamma(g) = q_g + u_g$ in the chosen based slice. For $0 \leq t \leq 1$, set

$$(95) \quad \gamma_t(g) = q_g + t u_g.$$

The parameter-separation argument is uniform in t , because $\|t u_g\|_{L_1^2} \leq C \lambda^{1/3}$ and the derivative of the modulation map remains $M_{\lambda, N} + o(1)$. Hence every γ_t is a compactified embedding. The construction is equivariant, γ_0 is the splicing map, and $\gamma_1 = \gamma$.

6. REVERSE GLUING AND NEIGHBORHOOD SURJECTIVITY

Continuity and injectivity do not imply that every nearby monopole is obtained by gluing. We prove the converse by first putting an arbitrary one-bubble sequence close to a preglued pair and then applying the same complement and inverse used in the construction.

Proposition 6.1 (strong splicing approximation). *Suppose that irreducible monopoles $[A_n, \Phi_n]$ converge in the Uhlenbeck topology to*

$$([B, \Psi], x) \in M_s \times X$$

with exactly one unit of instanton charge lost at x . There are coarse gluing data

$$g_n^0 = ([B, \Psi], x_n, \lambda_n, \rho_n), \quad x_n \rightarrow x, \quad \lambda_n \downarrow 0,$$

standard gluing bundle identifications, and determinant-one gauges for which

$$(96) \quad \|(A_n, \Phi_n) - q_{\lambda_n}(g_n^0)\|_{L_1^2} \rightarrow 0.$$

The relative frames ρ_n use the quotient convention in (25); no auxiliary tangent or bundle frame remains in the datum.

6.1. Extraction of the bubble. The monopole Weitzenböck formula gives a uniform L^∞ bound for Φ_n on the fixed compact perturbation family. Choose x_n and λ_n by a fixed fractional-energy centering and radius convention in a normal-coordinate ball about x . Pull back by $y \mapsto \exp_{x_n}(\lambda_n y)$ and choose Uhlenbeck gauges. The rescaled curvature equation has the form

$$(97) \quad F_{A_n^{\text{res}}}^+ = \lambda_n^2 \mathcal{Q}(\Phi_n^{\text{res}}) + \lambda_n^2 \mathcal{P}_n,$$

where $\mathcal{P}_n$ is bounded on each fixed normalized ball. The right side tends to zero. Uhlenbeck compactness and removal of the singularity at infinity therefore give a finite-energy ASD connection I on S^4 . The lost charge is one, so I has charge one. The centering and radius conventions make it centered and of unit scale; the framed charge-one classification identifies the remaining parameter with a based rotation $\rho_n \in \text{SO}(3)$. We choose this rotation by comparing the rescaled bubble gauge with the outer background gauge through the annular patching normalization below; this is equivalent to the framed classification and prevents a second, unrecorded frame from entering the gluing datum.

After a subsequence,

$$(98) \quad \begin{aligned} A_n^{\text{res}} &\longrightarrow I && \text{strongly in } L_1^2 \text{ on compact subsets of } \mathbb{R}^4, \\ (A_n, \Phi_n) &\longrightarrow (B, \Psi) && \text{strongly in } L_1^2 \text{ on compact subsets of } X \setminus \{x\}. \end{aligned}$$

The conformally rescaled spinor tends to zero on the bubble side.

On a conformal cylinder, the connection gauge-curvature block has tangential operator

$$(99) \quad L_0(\xi, b) = (-d_{S^3}^* b, -d_{S^3} \xi + *_{S^3} d_{S^3} b), \quad L = L_0 \oplus D_{S^3}.$$

For the unit round sphere, the exact-form spectrum of L_0 is $\{\pm\sqrt{k(k+2)} : k \geq 1\}$, its coexact spectrum is $\{\pm(k+1) : k \geq 1\}$, and its kernel consists exactly of the constant ξ modes. The spin Dirac spectrum is $\{\pm(k+3/2) : k \geq 0\}$. Hence, after removal of the constant gauge mode,

$$(100) \quad \text{spec } L \cap (-3/2, 3/2) = \emptyset.$$

Lemma 6.2 (fixed-annulus coupled estimate). *Let $D' \Subset D$ be one of the fixed round annulus pairs obtained by rescaling a dyadic annulus. Let (A, Φ) solve the monopole equation on D , and let (C, Ψ) solve it up to a defect $f \in L^p$, for a fixed $2 < p < 4$. Suppose both connections are in Coulomb gauge relative to the flat product connection and have scale-invariant $L^4 \oplus W^{1,2}$ norm below the Uhlenbeck constant. Then, for $z = (A - C, \Phi - \Psi)$,*

$$(101) \quad \|z\|_{W^{1,p}(D')} \leq C_p (\|z\|_{L^2(D)} + \|f\|_{L^p(D)}).$$

On an overlap, the based mismatch w between the actual and approximate transition maps satisfies the corresponding $W^{1,4} \cap W^{2,2} \cap W^{2,p}$ estimate with right-hand side given by the two neighboring z -norms and the defect.

Proof. Subtract the gauge, curvature, and Dirac equations. The result is a fixed first-order elliptic operator applied to z , with coefficients given by the two connection matrices and spinors, equal to f . Multiplication by a small L^4 connection coefficient maps $W^{1,p}$ to L^p , because

$$W^{1,p} \hookrightarrow L^{4p/(4-p)}, \quad \frac{1}{p} = \frac{1}{4} + \frac{4-p}{4p},$$

and can be absorbed. Interior elliptic regularity proves (101). Subtracting the two transition equations first gives the $W^{1,4}$ estimate for $w-1$; differentiating once and using (101) gives the $W^{2,2}$ and $W^{2,p}$ bounds on a smaller annulus. Since $p > 2$, the latter embeds in C^0 , placing w in one fixed logarithm chart. $\square$

Lemma 6.3 (two-sided cylinder estimate). *Suppose on $[j_-, j_+] \times S^3$ that $P_0 z = 0$ and*

$$(102) \quad (\partial_t + L)z = R(t)z + Q(z) + f,$$

where the operator norm of R and the Lipschitz norm of Q on each two-slab enlargement are sufficiently small. Let α_j be the scale-invariant $W^{1,2} \cap W^{1,p}$ norm of z there, for a fixed $2 < p < 4$, and let β_j be the corresponding $L^2 \cap L^p$ norm of f . There are $C > 0$ and $0 < \theta < 1$, independent of the cylinder length, such that

$$(103) \quad \alpha_j \leq C \left(\theta^{j-j_-} \alpha_{j_-} + \theta^{j_+-j} \alpha_{j_+} + \sum_{k=j_-}^{j_+} \theta^{|j-k|} \beta_k \right),$$

$$(104) \quad \sum_{j=j_-}^{j_+} \alpha_j \leq C \left(\alpha_{j_-} + \alpha_{j_+} + \sum_{j=j_-}^{j_+} \beta_j \right).$$

Proof. Let P_+ and P_- be the positive and negative spectral projections of L . Variation of constants, propagated from the left for P_+ and from the right for P_- , gives

$$\begin{aligned} P_+ z(t) &= e^{-(t-a)L} P_+ z(a) + \int_a^t e^{-(t-s)L} P_+ h(s) ds, \\ P_- z(t) &= e^{-(t-b)L} P_- z(b) - \int_t^b e^{-(t-s)L} P_- h(s) ds, \end{aligned}$$

where $h = Rz + Q(z) + f$. The gap (100) bounds the semigroups by $e^{-3|t-s|/2}$. Interior elliptic estimates give (103). The terms involving $Rz + Q(z)$ are the same geometric convolution with a small multiple of α_k ; its operator is bounded on both ℓ^1 and ℓ^∞ and can be absorbed. Summing the remaining geometric series proves (104). $\square$

Proposition 6.4 (family long-neck estimate). *Let (A_n, Φ_n) be either genuine monopoles compared with their coarse pregluings, or corrected configurations satisfying the full stabilized equation*

$$(105) \quad \mathfrak{S}(q_{g_n} + u_n) = J_{s,g_n}(b(q_n) + \eta_{s,n}) + J_{i,g_n} \eta_{i,n}, \quad d_{q_{g_n}}^{0,*} u_n = 0,$$

with $\|u_n\|_{L_1^2} + \|\eta_{s,n}\| + \|\eta_{i,n}\| \leq C \lambda_n^{1/3}$. Normalize at the exterior end. On $\Omega_n(d, R) = B(x_n, d) - B(x_n, R\lambda_n)$ there is one determinant-one gauge in which

$$(106) \quad \begin{aligned} &\|A_n - A_n^{\text{spl}}\|_{L^4} + \|\nabla_{A_n^{\text{spl}}}(A_n - A_n^{\text{spl}})\|_{L^2} + \|d_{A_n^{\text{spl}}}^*(A_n - A_n^{\text{spl}})\|_{L^{\sharp,2}} \\ &\leq C(\alpha_{n,-} + \alpha_{n,+} + d^{3/2} + d^2 + R^{-2} + \lambda_n^{1/3}). \end{aligned}$$

For genuine monopoles the last term may be omitted. The constant is independent of the number of dyadic annuli and of the point in the compact background family. The constant cylinder mode is retained at the inner end as the relative frame.

Proof. Put $r = e^{-t}$ and conformally change the Euclidean neck metric to $dt^2 + g_{S^3}$. A Euclidean spinor becomes $\Phi_{\text{cyl}} = r^{3/2}\Phi_E$. After radial and Coulomb gauge, the relative equation has principal part $\partial_t + L$. The normal-coordinate metric error, bounded background curvature, quadratic spinor term, and fixed perturbation have cylindrical size $O(r^2)$; the charge-one tail has size $O((\lambda_n/r)^2)$. On the at most two slabs meeting the outer spinor cutoff, conformal covariance gives the additional source

$$c(dt)(\partial_t \chi) r^{3/2} \Phi_E,$$

of size $O(r^{3/2})$. Thus the pregluing defect on a slab of radius r_j satisfies

$$(107) \quad \beta_j \leq C \left(r_j^2 + \frac{\lambda_n^2}{r_j^2} + \mathbf{1}_{\{|j-j_{\text{out}}| \leq 1\}} r_j^{3/2} \right),$$

and its sum is bounded by $C(d^{3/2} + d^2 + R^{-2})$.

For (105), the $b(q_n)$ terms cancel on subtraction. The background obstruction fields contribute $O(r_j^2 + r_j^{5/2})$, while the normalized charge-one adjoint-kernel spinor contributes $O(\lambda_n/r_j)$. Multiplication by the coefficient bound $C\lambda_n^{1/3}$ and geometric summation add at most $C\lambda_n^{1/3}$ to (104).

No-neck energy makes the coefficients in (102) uniformly small. Solve temporal gauge from the exterior end, removing the constant mode from z but retaining its inner endpoint. Lemma 6.3 gives the required length-independent ℓ^1 estimate. On neighboring fixed annuli, Lemma 6.2 gives, for the based transition mismatch w_j ,

$$\|w_j - 1\|_{W^{1,4} \cap W^{2,2} \cap W^{2,p}} \leq C(\alpha_j + \alpha_{j+1} + \beta_j + \beta_{j+1}).$$

Because $p > 2$, these maps lie in one logarithm chart. Extend their logarithms with fixed cutoffs, first over even and then over odd overlaps. The ℓ^1 estimate controls the critical divergence norm without a neck length factor and proves (106). The constant cocycle propagates uniquely from the exterior; its inner endpoint is the declared relative frame rather than a mode to be divided out. $\square$

Lemma 6.5 (compatible gauges on a long neck). *Let*

$$\Omega_n(d, R) = B(x_n, d) \setminus B(x_n, R\lambda_n),$$

and let A_n and C_n be determinant-one connections on the same splicing bundle over this annulus. Suppose that their total curvature energies on $\Omega_n(d, R)$ tend to zero in the iterated order $n \rightarrow \infty$, $d \downarrow 0$, and $R \rightarrow \infty$. Suppose also that, on fixed-ratio collars at the two ends, chosen gauges for A_n and C_n converge respectively to the same background connection and, after the declared rotation ρ_n , to the same framed bubble connection. Then there is one determinant-one gauge u_n on the whole neck, agreeing with these endpoint normalizations on smaller collars, such that

$$(108) \quad \begin{aligned} & \|u_n(A_n) - C_n\|_{L^4(\Omega_n(d, R))} + \|\nabla_{C_n}(u_n(A_n) - C_n)\|_{L^2(\Omega_n(d, R))} \\ & \leq C(\|F_{A_n}\|_{L^2(\Omega_n(d, R))} + \|F_{C_n}\|_{L^2(\Omega_n(d, R))}) + o_{n, d, R}(1), \end{aligned}$$

where C is independent of the number of dyadic annuli and the last term is the sum of the two endpoint discrepancies.

Proof. Use enlarged dyadic annuli D_j and concentric cores D'_j so that the D'_j cover the neck and the enlarged family has overlap bounded by an absolute constant. After rescaling each D_j to a fixed annulus, Uhlenbeck's small-curvature theorem [Uhl82] gives based Coulomb gauges a_j and c_j for A_n and C_n with

$$(109) \quad \|a_j\|_{L^4(D_j)} + \|\nabla a_j\|_{L^2(D_j)} + \|c_j\|_{L^4(D_j)} + \|\nabla c_j\|_{L^2(D_j)} \leq C e_j,$$

where

$$e_j^2 = \|F_{A_n}\|_{L^2(D_j)}^2 + \|F_{C_n}\|_{L^2(D_j)}^2.$$

Bounded overlap gives

$$(110) \quad \sum_j e_j^2 \leq C \int_{\Omega_n^+} (|F_{A_n}|^2 + |F_{C_n}|^2),$$

where Ω_n^+ enlarges only the two endpoint collars.

We record the overlap step because it is what prevents a logarithmic loss. If v_j is the transition map between two neighboring Coulomb gauges, then on the fixed-shape overlap

$$(111) \quad dv_j = v_j a_{j+1} - a_j v_j.$$

Poincaré's inequality applied to (111), followed by the polar projection of the average to the compact determinant-one group, produces a constant k_j such that

$$(112) \quad \|k_j^{-1} v_j - 1\|_{W^{1,4}} + \|k_j^{-1} v_j - 1\|_{W^{2,2}} \leq C(e_j + e_{j+1}).$$

The $W^{2,2}$ estimate follows by differentiating (111); its right-hand side is in L^2 by $L^4 \cdot L^4 \subset L^2$, and the Coulomb equations control the weak first derivatives of a_j and a_{j+1} . The same construction gives constants ℓ_j for the transitions of the c_j gauges. On a slightly smaller overlap, the gauge-fixed monopole equations for (a_j, Φ_n) and the explicit equation for the smooth spliced pair give, by interior elliptic regularity, a bound in $W^{2,p}$ for the transition maps for some fixed $p > 2$. Its norm relative to the constants is bounded by a modulus $\omega(e_j + e_{j+1})$ with $\omega(t) \rightarrow 0$. Hence $W^{2,p} \hookrightarrow C^0$ puts each based transition in one normal neighborhood of the identity. The group logarithm and cutoff extension used in the next paragraph are therefore well-defined. This bootstrap is used only to construct the gauges; the global quantitative estimate remains the critical square-sum estimate (110).

Fix the outer background normalization. Since the nerve of the annular cover is an interval, multiply successive local gauges by constants recursively so that the constant parts k_j and ℓ_j agree. The remaining overlap maps are the based maps in (112). The standard cutoff extension on the fixed overlap absorbs each such map into one of its two neighbors. Its $W^{1,4} \cap W^{2,2}$ norm is bounded by the right-hand side of (112). Applying the extensions on alternating overlaps first, and then on the remaining overlaps, gives global gauges. The resulting $L^4 \oplus W^{1,2}$ errors are estimated by the square sum, not by their ordinary sum. Thus (110) gives a constant independent of the length of the neck.

It remains to identify the normalization at the inner end. Cap the outer end by the background bundle and the inner end by the charge-one bubble bundle. The two capped bundles have the same second Chern class: their only clutching class is the unit charge already used in the splicing construction. Hence the product of the constant overlap transitions contains no additional $\pi_3(\mathrm{SO}(3))$ class. After the outer constant has been fixed, its inner constant is precisely the constant comparing the rescaled inner gauge with the centered framed instanton; by our convention following (97), this constant is ρ_n . Inserting ρ_n into C_n makes the endpoint constants agree. The endpoint convergence absorbs the two final collar errors into $o_{n,d,R}(1)$ and proves (108). $\square$

Lemma 6.6 (one-bubble no neck). *For the centers and scales above,*

$$(113) \quad \lim_{R \rightarrow \infty} \lim_{d \downarrow 0} \limsup_{n \rightarrow \infty} \int_{B(x_n, d) \setminus B(x_n, R\lambda_n)} |F_{A_n}|^2 = 0.$$

After the bundle identifications determined by ρ_n , there are gauges on these necks in which the actual connection and the spliced background–instanton connection satisfy

$$(114) \quad \|A_n - A_n^{\text{spl}}\|_{L^4} + \|\nabla_{A_n^{\text{spl}}}(A_n - A_n^{\text{spl}})\|_{L^2} \rightarrow 0$$

in the iterated order $n \rightarrow \infty$, $d \downarrow 0$, and $R \rightarrow \infty$.

Proof. The curvature measures converge to

$$|F_B|^2 dV + 8\pi^2 \delta_x.$$

For fixed d , the region outside $B(x_n, d)$ accounts for the background energy. After rescaling, $B(x_n, R\lambda_n)$ accounts for the energy of I on $B(0, R)$. Taking the three limits in the displayed order, the background energy in the shrinking ball and the instanton tail both vanish. There is no second concentration point, so (113) follows.

Cover the neck by dyadic annuli. Once their curvature energy is below the Uhlenbeck threshold, the general small-curvature Coulomb-gauge estimate [Uhl82] gives

$$(115) \quad \|a\|_{L^4} + \|\nabla a\|_{L^2} \leq C\|F_A\|_{L^2}.$$

This is the estimate for a general connection, not an ASD annular decay theorem. It applies here because the full neck energy tends to zero. The monopole equation and the uniform spinor bound also give

$$(116) \quad \|F_{A_n}^+\|_{L^2(B(x_n, d))} \leq Cd^2.$$

The annuli have bounded overlap, so the squares of (115) sum to a constant times the total neck energy; no factor equal to the number of annuli occurs. Apply the same estimate to the spliced connection, whose neck energy is small by bounded background curvature and instanton-tail decay. The two-sided convergence in (98) supplies the endpoint hypotheses of Lemma 6.5; the spliced connection uses the same outer background gauge and the same inner rotation ρ_n . Apply (108) and then the three limits. This proves (114) in one global determinant-one gauge. $\square$

Proof of Proposition 6.1. Given $\epsilon > 0$, first choose R so that the L_1^2 tail of I outside $B(0, R)$ is smaller than ϵ . Next choose d so that the background has small local energy in $B(x, 2d)$ and the no-neck quantity is smaller than ϵ . For all large n , the first convergence in (98) controls $B(x_n, R\lambda_n)$, Lemma 6.6 controls the intermediate annulus, and the second convergence controls $X \setminus B(x_n, d)$.

The spinor has no missing neck term. Its pointwise bound gives

$$(117) \quad \|\Phi_n\|_{L^4(B(x_n, 2d))} = O(d),$$

and the Dirac equation with an interior covariant Caccioppoli estimate gives

$$(118) \quad \|\nabla_{A_n} \Phi_n\|_{L^2(B(x_n, d))} \leq Cd.$$

Indeed, the Weitzenböck formula bounds the square of the left side by the cutoff contribution, bounded zeroth-order terms, and

$$\int_{B(x_n, 2d)} |F_{A_n}| |\Phi_n|^2 \leq \|\Phi_n\|_{L^\infty}^2 \|F_{A_n}\|_{L^2} \text{vol}(B(x_n, 2d))^{1/2} = O(d^2).$$

The cutoff background spinor satisfies the same estimates. Patching the three comparisons with the pregluing profiles yields

$$\limsup_{n \rightarrow \infty} \|(A_n, \Phi_n) - q_{\lambda_n}(g_n^0)\|_{L_1^2} \leq C\epsilon.$$

Letting ϵ tend to zero proves (96). $\square$

6.2. Exact modulation and complemented uniqueness. Apply Lemma 5.2 to the configurations in Proposition 6.1. After based determinant-one gauge, there are unique exact gluing data $g_n = (q_n, \hat{x}_n, \hat{\lambda}_n, \hat{\rho}_n)$ and differences u_n such that

$$(119) \quad (A_n, \Phi_n) = q_{\hat{\lambda}_n}(g_n) + u_n, \quad d_{q_{\hat{\lambda}_n}}^{0,*} u_n = 0, \quad R_{g_n} u_n = 0, \quad \|u_n\|_{L_1^2} \rightarrow 0.$$

The exact parameters are close to the coarse ones in (54); in particular, q_n tends to the reducible background and $b(q_n) \rightarrow 0$.

Set

$$(120) \quad H_n = (u_n, -b(q_n), 0).$$

The complement condition depends only on u_n , so H_n lies in $\mathbb{H}_{\hat{\lambda}_n, N}^{\text{cmp}}$. Since (A_n, Φ_n) is a monopole and is in the based slice,

$$(121) \quad \hat{\mathcal{F}}_{g_n}(H_n) = 0.$$

Let $e_n = \hat{\mathcal{F}}_{g_n}(0)$. The inverse and quadratic estimates give

$$\|H_n\| \leq C\|e_n\| + C\|H_n\|^2 \leq C\hat{\lambda}_n^{1/3} + C\|H_n\|^2.$$

The qualitative convergence in (119) absorbs the quadratic term, and hence

$$(122) \quad \|H_n\| \leq 2C\hat{\lambda}_n^{1/3}.$$

Thus H_n lies in the same contraction ball as the correction in Proposition 4.2. Uniqueness of that fixed point gives

$$(123) \quad (u_n, -b(q_n), 0) = (u_{g_n}, \eta_{s, g_n}, \eta_{i, g_n}).$$

It follows that $[A_n, \Phi_n] = \gamma(g_n)$ and $\chi(g_n) = 0$.

Proposition 6.7 (neighborhood surjectivity). *There is an Uhlenbeck-open neighborhood $\mathcal{U}$ of $M_5 \times X$ such that*

$$(124) \quad \mathcal{U} \cap \overline{\mathcal{M}}_v^{*,0} = \mathcal{U} \cap \gamma(\chi^{-1}(0)).$$

Consequently, γ restricts to a homeomorphism from $\chi^{-1}(0)$ onto an Uhlenbeck-open neighborhood of the ideal stratum, after restricting the virtual neighborhood over its outer boundary.

Proof. The gluing construction gives the inclusion from right to left. If the reverse inclusion failed in every neighborhood, there would be genuine monopoles outside $\gamma(\chi^{-1}(0))$ converging to the compact ideal stratum. A subsequence converges to one point of that stratum, and the argument above places every sufficiently late member in $\gamma(\chi^{-1}(0))$, a contradiction. The inverse is continuous because γ is a compactified embedding by Corollary 5.5. $\square$

7. THE OBSTRUCTION CONE AND ORIENTATIONS

7.1. The compactified section. The background summand $\overline{\Upsilon}^s$ is the pullback of the fixed complex bundle $\Xi \rightarrow N_{\psi,s}$. The charge-one Dirac operator has the kernel and cokernel in (24). At positive scale its obstruction line is

$$(0, \delta] \times \mathrm{SU}(2) \times_{\{\pm 1\}} \mathbb{C} \longrightarrow (0, \delta] \times \mathrm{SO}(3).$$

Its compactification is the cone map (29); the whole fiber, not merely its norm, is identified with the vertex at zero scale. There are no partition-overlap maps at level one.

At positive scale, define χ by (78); at zero scale, set

$$(125) \quad \chi(q, x, 0) = (b(q), 0).$$

The contraction is equivariant, because the nonlinear map, cutoffs, transports, and inverse are equivariant and its fixed point is unique. The two common presentations of the residual circle action should not be mixed. Before taking the effective circle, the diagonal presentation has weight two on the normal and background-obstruction factors. After passage to the effective action, the corresponding Euler class is

$$(126) \quad e(\overline{\Upsilon}^s/S^1) = (-\nu)^{r^\Xi}.$$

The instanton line retains the nontrivial $\mathrm{SU}(2)/\{\pm 1\}$ framing twist, so its Euler class is computed from its tensor square in Section 8 rather than copied from (126).

Proposition 7.1 (compactified obstruction section). *The two components of χ are smooth at positive scale. The background component extends continuously over the compactification and smoothly on each zero-scale stratum. The instanton component is a continuous section of the cone pseudo-bundle. With the metric induced by $J_{s,g} \oplus J_{i,g}$, the norm of χ is continuous along the section.*

Proof. Positive-scale smoothness follows from Proposition 4.2. At zero scale, its estimate gives

$$\|\eta_{s,g}\| + \|\eta_{i,g}\| \leq C\lambda^{1/3}.$$

Thus χ_s tends to $b(q)$ and χ_i tends to the cone vertex. For $g_n \rightarrow (q, x, 0)$, the Gram estimate in Lemma 3.3 gives

$$\|J_{s,g_n}\chi_s(g_n) + J_{i,g_n}\chi_i(g_n)\|_{L^2}^2 = \|b(q)\|^2 + o(1),$$

which is the asserted norm continuity. This does not extend the ambient instanton line as a constant-rank bundle; only the section tends to its collapsed fiber. $\square$

We next compare the global reduction with the local small-eigenvalue reduction of [FL99, Section 8]. Three objects must be kept distinct. On a relatively compact positive-scale chart U^+ on which a cutoff μ avoids the spectrum, let

$$(127) \quad E_{\mathrm{glob},g} = \Xi_q \oplus K_I, \quad E_{\mathrm{loc},g} = \ker d_q^{1,*} \oplus K_I, \quad E_{\mathrm{spec},g} = \mathrm{Ran} \Pi_{q_g,\mu}.$$

The published cut-and-project and synthesis estimates [FL99, Propositions 8.19 and 8.26] identify E_{loc} with E_{spec} on a constant-rank chart: their two composites are $I + o(1)$. They also transport the ordered orientation local system of $\ker d_q^{1,*} \oplus K_I$ to E_{spec} , compatibly on overlaps. There is generally no isomorphism between E_{glob} and E_{loc} , since the global augmentation may contain extra stabilizing directions.

Proposition 7.2 (stable comparison of reductions). *On every positive-scale constant-rank germ of the monopole zero set, the global and spectral reductions admit a common equivariant finite-dimensional enlargement. Their zero loci agree, the comparisons satisfy the cocycle condition up to controlled equivariant homotopy, and the induced determinant orientation and relative Euler–Thom pairing agree. No equality of the two ranks and no spectral bundle at zero scale is asserted.*

Proof. For a smooth Fredholm section F with two surjective augmentations $i_j : E_j \rightarrow Y$, form

$$\mathcal{V}_{01} = \{(x, e_0, e_1) : F(x) - i_0 e_0 - i_1 e_1 = 0\}.$$

The two reductions embed by setting the other coefficient equal to zero. At a point of the first reduction, projection to e_1 identifies its normal bundle in $\mathcal{V}_{01}$ with E_1 : surjectivity of $DF \oplus (-i_0)$ supplies a lift of every prescribed e_1 . The analogous statement holds with the indices reversed. Thus both obstruction zero loci are canonically $F^{-1}(0)$.

Apply this construction with $E_0 = E_{\text{glob}}$ and $E_1 = E_{\text{spec}}$. The first augmented derivative is surjective by Theorem 3.5. On $E_{\text{spec}}^\perp$, the second has right inverse

$$d_{q_g}^{1,*} (d_{q_g}^1 d_{q_g}^{1,*})^{-1},$$

bounded by the spectral gap at μ . Lemma 3.6 and the Riesz formula for $\Pi_{q_g, \mu}$ give controlled tubular maps on a common germ. On overlaps, order all spectral factors by chart label and use literal zero-coordinate embeddings; associativity of direct sum gives the cocycle before tubular choices, and nearby invariant tubular choices are homotopic through their graph maps.

Along the first embedding,

$$\det T\mathcal{V}_{01} \cong \det T\mathcal{V}_0 \otimes \det E_1.$$

The same E_1 occurs in the common obstruction bundle, so its determinant line cancels its dual. Along the other embedding the fixed order contributes the Koszul factor $(-1)^{\text{rank}_{\mathbb{R}} E_0 \text{rank}_{\mathbb{R}} E_1}$; it is $+1$ because $E_0 = \Xi \oplus K_I$ has even real rank. If E_1 is not globally oriented, its orientation local system is carried by both factors and cancels canonically. The identical cancellation holds in relative equivariant cohomology: the Thom class of the added normal E_1 is the Euler factor of the added obstruction E_1 . Hence stabilization changes neither the oriented zero locus nor the relative Euler–Thom pairing. $\square$

7.2. The exact transversality statement. At a positive-scale zero, the extended equation is

$$(128) \quad \mathfrak{S}(\gamma(g)) = J_g \chi(g).$$

Let Q_g be projection onto the obstruction image. The contraction parametrizes the local virtual manifold $(1 - Q_g)\mathfrak{S} = 0$, and the complemented inverse identifies its tangent space with $D\gamma_g(T_g \mathcal{G}^+)$. Differentiating (128) at a zero gives

$$(129) \quad D\mathfrak{S}_{\gamma(g)} D\gamma_g = J_g D\chi_g.$$

Proposition 7.3 (stratumwise transversality). *For a generic perturbation at which the irreducible monopole section is transverse, the total section $\chi_s \oplus \chi_i$ is transverse on $\mathcal{G}^+$. On $(N_{\mathfrak{t}, s} - M_s) \times X$, the surviving background component $\chi_s = b$ is transverse. No transversality holds or is asserted on $M_s \times X$ when $r_\Xi > 0$.*

Proof. Given $e \in \Upsilon_g$ at a positive-scale zero, surjectivity of $D\mathfrak{S}_{\gamma(g)}$ supplies v with $D\mathfrak{S}_{\gamma(g)}v = J_g e$. Its projected equation component vanishes, so v is tangent to the virtual manifold and equals $D\gamma_g \dot{g}$ for some $\dot{g}$. Equation (129) and injectivity of J_g give $D\chi_g \dot{g} = e$.

On the middle stratum, the instanton fiber has collapsed and the background finite-dimensional reduction makes b transverse; the X factor is passive. On the lowest stratum b is identically zero. An identically zero section of a positive-rank bundle is not transverse. This is why [FL02, Theorem 3.8 and Section 5.1] asserts precisely the first two conclusions and disclaims the third. $\square$

Accordingly, the all-strata formulation in [FL18, Hypothesis 7.8.1(2)] must be read using the coarser stratification there. For the fine decomposition (27), the relative Euler construction uses Proposition 7.3 and does not require transversality on the lowest stratum.

7.3. Determinant-line excision. For a real Fredholm operator F , use

$$\det F = \Lambda^{\max} \ker F \otimes (\Lambda^{\max} \operatorname{coker} F)^*.$$

Direct-sum factors are ordered as written, and complex vector spaces carry their complex orientations. For $q \in N(\varepsilon)$, write $p = \pi_N(q) \in M_{\mathfrak{s}}$. Orient $T_p M_{\mathfrak{s}}$ using the homology orientation Ω . Orient the centered framed instanton parameters by

$$(130) \quad (\partial_\lambda, i, j, k) \quad \text{on} \quad \mathbb{R} \oplus \mathfrak{so}(3) = \mathbb{H}.$$

We use the real determinant-line excision principle of [Don87, Section 3]; see also [DK90, Section 7.1]. In the form needed here, suppose two ordered pairs of real first-order elliptic operators are obtained from one another by exchanging compact pieces, and that their bundles, principal symbols, and operators agree on product collars after a homotopy of zero-order terms. There is then a natural isomorphism

$$(131) \quad \text{Exc} : \det D_1 \otimes \det D_2 \longrightarrow \det D'_1 \otimes \det D'_2.$$

It is continuous for families, compatible with composition and direct sum, and is the identity when the exchanged data are identical. These normalizations determine its effect on orientations. This is the excision principle used to define $O^{\text{asd}}(\Omega, w)$ in [FL01b, Sections 2.1–2.2].

We apply (131) to the augmented operators, for which the parametrix supplies all required uniform Fredholm control. Replace the splicing annulus by a cylinder $[-T, T] \times S^3$ and denote the resulting operator by $\widehat{\mathcal{D}}_{g,T}$. On the central third of the cylinder, homotope the coefficients to the product coefficients and remove the zero-order terms. On the two remaining thirds use the same transports as in (62). The commutator and coefficient estimates of Section 3 show that, after T is large, this entire homotopy has a uniformly invertible restriction to the model-coordinate complement. Thus its determinant line is transported without a spectral crossing on that complement.

Cut at the middle of the product cylinder and cap the two pieces by the fixed background and bubble models. Excision gives

$$(132) \quad \det \widehat{\mathcal{D}}_q^X \otimes \det \widehat{B}_I \cong \det \widehat{\mathcal{D}}_{q_g} \otimes \det D_{\text{ref}}.$$

Here D_{ref} is the fixed based product operator obtained by gluing the two discarded flat caps. Choose its fixed finite-dimensional stabilization so that it is invertible. Its determinant is then canonically identified with $\mathbb{R}$ by $1 \in \Lambda^0(0)$, and this factor is suppressed below. Equivalently, the isomorphism after suppressing it is the determinant of the kernel–cokernel exact

sequence constructed by the patched parametrix. Because the three augmented operators are surjective, that exact sequence reduces to the kernel gluing isomorphism

$$(133) \quad \ker \widehat{\mathcal{D}}_{q_g} \cong \ker \widehat{\mathcal{D}}_q^X \oplus \ker \widehat{B}_I.$$

The synthesis and retraction maps in (55)–(56) represent this isomorphism; its matrix is $I + o(1)$ and therefore has positive determinant.

The ordered model kernels and finite-dimensional stabilizations give

$$(134) \quad \begin{aligned} \det \widehat{\mathcal{D}}_q^X &= \det N_p \otimes \det T_p M_s, \\ \det \widehat{B}_I &= \det T_I M_1^s(S^4, \delta), \\ \det \widehat{\mathcal{D}}_{q_g} &= \det \mathcal{D}_{q_g} \otimes \det \Xi_p \otimes \det K_I. \end{aligned}$$

The last identity is the determinant exact sequence for adding the finite-dimensional domain $\Xi_p \oplus K_I$ with the ordered map $-(J_{s,\lambda} \oplus J_{i,\lambda})$. Its minus sign has positive determinant because the real stabilization dimension is $2r_\Xi + 2$. Combining (132)–(134) and putting the obstruction determinants on the dual side yields

$$(135) \quad \begin{aligned} \det \mathcal{D}_{q_g} &\cong \det N_p \otimes \det T_p M_s \otimes \det T_I M_1^{s,\natural}(S^4, \delta) \otimes \det T_x X \\ &\otimes (\det \Xi_p)^* \otimes (\det K_I)^*. \end{aligned}$$

Here the centered bubble factor contains scale and relative rotation; its four translations are identified orientation-preservingly with $T_x X$ by the chosen oriented tangent frame. Moving the four-dimensional translation factor past the four-dimensional scale–rotation factor contributes no sign.

The splitting (135) has no orientation monodromy around a loop in M_s . The tangent bundle is oriented by Ω , the bundles N and Ξ have their complex orientations, the instanton modes have the fixed quaternionic orientation, TX is oriented, and K_I is a complex line. Possible spectral flow of an associated self-adjoint family changes neither this real orientation nor the global stabilization.

The tangent- and adjoint-frame determinants introduced before taking the diagonal $\mathrm{SO}(4) \times \mathrm{SO}(3)$ quotient cancel between the ordered tangent exact sequences for the framed parameter space and for the orbit directions. Both group dimensions are moved past identical factors, so the two Koszul signs cancel. The remaining relative-frame factor is the $\mathrm{SO}(3)$ already contained in $M_1^{s,\natural}$.

There is also no orientation monodromy around this relative-frame factor. The generator of $\pi_1(\mathrm{SO}(3))$ lifts to a path in $\mathrm{SU}(2)$ from 1 to -1 . Equivariance of (131) transports the determinant along this lift. At the endpoint, -1 acts trivially on adjoint connection directions, by complex multiplication by -1 on each complex spinor or obstruction factor, and by -1 on the four-dimensional quaternionic parameter space. Each real determinant is $+1$. Hence the excision orientation descends around the nontrivial loop in $\mathrm{SO}(3)$.

Finally, the path $q_g + tu_g$ transports $\det \mathcal{D}_{q_g}$ to $\det \mathcal{D}_{\gamma(g)}$. Intermediate configurations need not solve the equations. Replacing the published obstruction profiles by the matched profiles is an isotopy through embeddings, and the model-coordinate Gram map remains positive. The normalization discrepancy in the spinor-bilinear term is removed by a zero-order homotopy through positive coefficients. Thus (135), with the same ordered factors, holds for the corrected operator and is the convention of [FL02, Section 3.9].

Proposition 7.4 (orientation of the corrected chart). *At a positive-scale zero of χ , there is a canonical orientation isomorphism*

$$(136) \quad \det \mathcal{D}_{\gamma(g)} \cong \det T_g \mathcal{G}^+ \otimes (\det \Upsilon_g)^*.$$

Under this isomorphism, $D\gamma$ identifies the transverse-zero-locus orientation of $\chi^{-1}(0)$ with the orientation $O^{\mathrm{asd}}(\Omega, c_1(L))$ of the actual monopole moduli space.

Proof. The tangent line of the virtual domain is the product of the first four factors in (135), while $\Upsilon_g = \Xi_p \oplus K_I$. This gives (136) as a line isomorphism. To identify it with the finite-dimensional reduction, differentiate (128) at a zero. The derivative of J_g drops out, and

$$\dot{g} \mapsto (D\gamma_g \dot{g}, D\chi_g \dot{g})$$

identifies $T_g \mathcal{G}^+$ with the kernel of the stabilized operator. It is homotopic under the model retraction to the identity, with positive Gram determinant. The ordering $K_I \oplus \Xi_p$ used in one source formula may be interchanged with $\Xi_p \oplus K_I$ without sign because both have even real dimension. Similarly, the analytic stabilization is $-J_g$, but $\dim_{\mathbb{R}} \Upsilon_g = 2r_{\Xi} + 2$ is even.

Proposition 7.3 gives the exact sequence

$$0 \longrightarrow T_g \chi^{-1}(0) \longrightarrow T_g \mathcal{G}^+ \xrightarrow{D\chi_g} \Upsilon_g \longrightarrow 0.$$

Its determinant is the right side of (136) with the same ordered factors. The claim follows. $\square$

7.4. The residual circle and the link boundary. Let $\mathcal{Z}^+ = \chi^{-1}(0) \cap \mathcal{G}^+$. On the free circle locus define

$$(137) \quad \lambda_{\mathcal{Z}} = v_{S^1} \wedge \tilde{\lambda}_{\mathcal{Z}/S^1}.$$

The weight-two presentation changes the positive speed of an orbit but not its orientation. Let $\vec{r}$ point outward from the deleted neighborhood of the ideal stratum. The boundary of its complement has

$$\lambda_{\mathcal{Z}/S^1} = -\vec{r} \wedge \lambda_L.$$

Together with (137), this proves

$$(138) \quad \lambda_{\mathcal{M}} = -v_{S^1} \wedge \vec{r} \wedge \tilde{\lambda}_L,$$

which is the convention in [FL02, Equation (3.81)]. The sign can also be read in one complex fiber: moving the positive orbit vector $i\vec{r}$ past the outward radial vector contributes the displayed minus sign.

The compactified virtual link has a background-normal face and an instanton-scale face. Their common edge receives opposite orientations, because exchanging the two outward normals contributes a minus sign. Thus the faces form an oriented relative cycle. This statement concerns the smooth positive-scale zero set and the relative Euler cycle; it does not reinterpret the lowest stratum as a transverse zero set.

Finally, if w is an integral lift of $w_2(\mathfrak{t}')$, the orientation comparison of [FL01b] is

$$(139) \quad O^{\mathrm{asd}}(\Omega, w) = (-1)^{(w - c_1(L))^2/4} O^{\mathrm{asd}}(\Omega, c_1(L)).$$

The exponent is integral. The circle and boundary constructions are linear in the ambient orientation, so the same factor relates the corresponding link orientations. This completes the orientation assertion in Theorem 2.1.

8. THE LEVEL-ONE PAIRING

We now apply the family neighborhood theorem to the published level-one pairing. The neighborhood theorem is used to identify the actual and virtual links, pull back geometric representatives, carry the universal bundle through an equivariant homotopy to the splicing map, represent the smooth zero locus by a relative Euler class, and preserve signed intersections. Corollary 5.5, Proposition 6.7, Proposition 7.1, and Proposition 7.4 supply these inputs. The dimension argument of [FL02, Lemmas 3.9 and 5.4] puts the counted intersection in the positive-scale smooth locus and away from the common edge. The universal-bundle calculation, Thom and Segre reduction, charge-one fiber evaluations, and Jacobi identity are retained as published topological inputs; the contribution here is to discharge their deferred analytic neighborhood hypothesis.

8.1. Pullbacks and Euler classes. Let

$$(140) \quad \nu = c_1(\bar{\mathcal{G}}^{\text{vir},*} \longrightarrow \bar{\mathcal{G}}^{\text{vir},*}/S^1)$$

for the effective circle. The line defining μ_c on the monopole configuration quotient has weight -2 , while the action on the gluing domain before effectivization has twice the angular parameter. Hence

$$(141) \quad \gamma^* \mu_c = -\nu.$$

Away from the incidence diagonal, the pulled-back universal $SO(3)$ bundle splits as $\mathbb{R} \oplus \bar{L}^{\text{vir}}$, where

$$(142) \quad c_1(\bar{L}^{\text{vir}}) = 2\pi_s^* \mu_s - \nu + \pi_{X,2}^* c_1(L), \quad c_1(L) = -\beta.$$

The equivariant isotopy (95) identifies the corrected and splicing pullbacks. Insertion of one instanton changes the Pontrjagin class by the incidence diagonal, so

$$(143) \quad \begin{aligned} (\gamma \times \text{id}_X)^* p_1(\mathbb{F}_{\nu}) &= (2\pi_s^* \mu_s - \nu + \pi_{X,2}^* c_1(L))^2 \\ &\quad - 4(\pi_X \times \text{id}_X)^* \text{PD}[\Delta]. \end{aligned}$$

The coefficient -4 is fixed by a chain meeting the diagonal once. Since $\mu_p(\alpha) = -\frac{1}{4}p_1(\mathbb{F}_{\nu})/\alpha$, slant product with a point and a surface gives

$$(144) \quad \begin{aligned} \gamma^* \mu_p(x) &= -\frac{1}{4}(2\mu_s - \nu)^2 + \pi_X^* \text{PD}[x], \\ \gamma^* \mu_p(h) &= \frac{1}{2}\langle \beta, h \rangle (2\mu_s - \nu) + \pi_X^* \text{PD}[h]. \end{aligned}$$

These are the formulas of [FL02, Lemma 4.10]; the second summand in each line is the diagonal correction.

The background Euler class is (126). For the instanton factor, the square of the framing line loses the central twist:

$$(\text{SU}(2) \times_{\{\pm 1\}} \mathbb{C})^{\otimes 2} \cong \text{SO}(3) \times \mathbb{C}.$$

Accounting for the central and residual circle weights gives

$$(145) \quad c_1((\Upsilon^i/S^1)^{\otimes 2}) = \pi_X^* c_1(\mathfrak{t}') - \nu.$$

Thus, in rational cohomology,

$$(146) \quad e(\Upsilon^i/S^1) = \frac{1}{2}(\pi_X^* c_1(\mathfrak{t}') - \nu).$$

The factor $1/2$ is the square-root factor of the framed associated line; it has no relation to the derivative-normalization homotopy discussed before (135).

8.2. Relative Euler reduction. On the positive-scale virtual link, the section determines a relative Euler class. The geometric-representative cocycle is supported away from the lower-level zero set. The homotopy

$$(147) \quad \chi_s \oplus \chi_i \mapsto \chi_s \oplus t\chi_i, \quad 0 \leq t \leq 1,$$

does not vanish off $\chi_s^{-1}(0)$. At $t = 0$ the class factors as

$$(148) \quad e(\Upsilon/S^1, \chi_s) = e(\overline{\Upsilon}^s/S^1, \chi_s) \smile e(\Upsilon^i/S^1).$$

This is precisely where one needs continuity of the background section and the cone topology, but not transversality of the lowest stratum. The signed actual-link count is therefore

$$(149) \quad \left\langle \overline{\mu}_p(z) \smile \overline{\mu}_c^\eta \smile \overline{e}_s \smile \overline{e}_i, [\overline{L}^{\text{vir}}] \right\rangle.$$

Write

$$H = \pi_X^* \text{PD}[h], \quad X_0 = \pi_X^* \text{PD}[x], \quad c = \pi_X^* c_1(\mathfrak{t}').$$

Substitution of (141), (144), (126), and (146) into (149) gives

$$(150) \quad \begin{aligned} \mathcal{P} = & \left\langle \left[\frac{1}{2} B_h(2\mu_{\mathfrak{s}} - \nu) + H \right]^n \right. \\ & \smile \left[-\frac{1}{4}(2\mu_{\mathfrak{s}} - \nu)^2 + X_0 \right]^m (-\nu)^{\eta+r_{\Xi}} \\ & \left. \smile \frac{1}{2}(c - \nu), [\overline{L}^{\text{vir}}] \right\rangle. \end{aligned}$$

The dimension relation is

$$(151) \quad \delta + \eta = d + r_N - r_{\Xi} + 2.$$

The equivariant Thom division of [FL02, Propositions 5.20–5.21] expresses this pairing over

$$M_{\mathfrak{s}} \times \partial \overline{\text{Gl}}_{\ell'}(\delta)/S^1$$

as a finite sum involving the Segre classes of N . The normal-bundle calculation [FL02, Lemma 2.4] gives

$$(152) \quad s_j(N) = S_j \mu_{\mathfrak{s}}^j,$$

where S_j is the binomial coefficient in [FL02, Equation (6.14)]. On this product base, [FL02, Lemma 5.23] gives

$$(153) \quad \nu = \nu_{\mathfrak{t}} + 2\mu_{\mathfrak{s}}.$$

Every nonzero term then has base factor

$$(154) \quad \left\langle \mu_{\mathfrak{s}}^d, [M_{\mathfrak{s}}] \right\rangle = \text{SW}_X(\mathfrak{s}).$$

In particular, positive-dimensional base classes have not been discarded; the Segre calculation combines all of them into the single pairing in (154).

Put

$$(155) \quad A_J = \eta - d + 1, \quad B_J = \frac{1}{2}(2\delta - d_a(\mathfrak{t}') - d_{\mathfrak{s}}) - \frac{1}{4}(\chi + \sigma),$$

and set

$$(156) \quad J_d = P_d^{A_J, B_J}(0), \quad J'_d = P_d^{A_J-1, B_J+1}(0),$$

using (11). The finite Segre sum is $2^d J_d$, except in the term containing Hc , where it is $2^d J'_d$; this is the Jacobi identity of [FL02, Lemma 6.2].

Lemma 8.1 (published charge-one link integrals). *Let ν_t denote the effective circle class on the charge-one instanton link. With the orientation in (138),*

$$\begin{aligned}
 \langle \nu_t X_0, [\partial \overline{\text{Gl}}/S^1] \rangle &= 2, & \langle \nu_t^2 H, [\partial \overline{\text{Gl}}/S^1] \rangle &= -4B_h, \\
 \langle \nu_t Hc, [\partial \overline{\text{Gl}}/S^1] \rangle &= 2\langle c_1(\mathfrak{t}'), h \rangle, & \langle \nu_t^3, [\partial \overline{\text{Gl}}/S^1] \rangle &= 6\beta^2 + 2c_1^2(X), \\
 (157) \quad \langle \nu_t^2 c, [\partial \overline{\text{Gl}}/S^1] \rangle &= -4\beta \cdot c_1(\mathfrak{t}').
 \end{aligned}$$

Proof. These numerical evaluations are cohomological inputs from [FL02, Lemma 5.24 and Equation (5.77)]; they are not reproved here. We recall enough of that calculation to fix their normalization. On a finite cover of X , the framed charge-one gluing bundle is covered with degree -2 by the boundary of a rank-two Hermitian bundle F . The induced map of projectivizations has degree -1 , while ν_t pulls back to twice the hyperplane class. The projective-bundle pushforward

$$\pi_*(h^{1+j}) = s_j(F)$$

therefore gives $\langle \nu_t X_0 \rangle = 2$ and, from the published Segre classes of F , the second and fourth values in (157). Taking $h = \text{PD}(c)$ in the second value gives the fifth. Finally, Poincare duality in Equation (5.77) gives

$$\langle \nu_t Hc \rangle = \langle \nu_t X_0 \rangle \langle c_1(\mathfrak{t}'), h \rangle,$$

which is the third value. Since $c_1(L) = -\beta$, these conventions agree with (141) and (138). $\square$

Combining the base reduction with Lemma 8.1 gives the following polynomial form, which does not divide by a Jacobi value and therefore remains defined when $J_d = 0$.

Theorem 8.2 (family level-one pairing). *Under the assumptions preceding Theorem B,*

$$\begin{aligned}
 &\#(\overline{V}(h^{\delta-2m}x^m) \cap \overline{W}^\eta \cap \overline{L}_{\mathfrak{t}',\mathfrak{s}}) \\
 &= (-1)^{m+1+d} 2^{d-\delta} \left\langle \mu_{\mathfrak{s}}^d, [M_{\mathfrak{s}}] \right\rangle \\
 &\quad \times [J_d(a_0 B_h^n + a_1 B_h^{n-2} Q_X(h, h)) + 2n J_d' B_h^{n-1} \langle c_1(\mathfrak{t}'), h \rangle],
 \end{aligned}$$

where terms with negative powers of B_h are omitted and

$$a_0 = 3\beta^2 + c_1^2(X) + 2\beta \cdot c_1(\mathfrak{t}') + 4n - 4m, \quad a_1 = 4 \binom{n}{2}.$$

Proof. Equation (150), equivariant Thom division, (152), and (153) reduce the base degree to (154). The Jacobi identity gives $2^d J_d$ and $2^d J_d'$ in the two indicated groups of terms. The five charge-one evaluations in Lemma 8.1 give respectively the $-4m$, $4n$, $2\beta \cdot c_1(\mathfrak{t}')$, $c_1^2(X)$, and $Q_X(h, h)$ contributions. Collecting them gives a_0 and a_1 . The sign and power of two are $(-1)^{m+1+d} 2^{d-\delta}$, as in [FL02, Equations (6.15)–(6.18)]; Proposition 7.4 introduces no further sign. This proves the formula. $\square$

Corollary 8.3 (zero-dimensional specialization). *If $d_{\mathfrak{s}} = 0$, then $J_0 = J_0' = 1$ and*

$$\begin{aligned}
 &\#(\overline{V}(h^{\delta-2m}x^m) \cap \overline{W}^\eta \cap \overline{L}_{\mathfrak{t}',\mathfrak{s}}) \\
 &= (-1)^{m+1} 2^{-\delta} \text{SW}_X(\mathfrak{s}) (a_0 B_h^n + b_0 B_h^{n-1} \langle c_1(\mathfrak{t}'), h \rangle + a_1 B_h^{n-2} Q_X(h, h)),
 \end{aligned}$$

where $b_0 = 2n$ and a_0, a_1 are as in Theorem 8.2.

8.3. Blow-up and the degree consequence. The only blow-up specialization needed below has no hidden change in a_0 . On $\tilde{X} = X \# \overline{\mathbb{CP}}^2$, let e be the exceptional class and

$$c_1(\mathfrak{s}^\pm) = c_1(\mathfrak{s}) \pm \text{PD}[e], \quad \text{SW}_{\tilde{X}}(\mathfrak{s}^\pm) = \text{SW}_X(\mathfrak{s}).$$

For $\tilde{\mathfrak{t}}'$ one has $c_1(\tilde{\mathfrak{t}}') = c_1(\mathfrak{t}')$ and $p_1(\tilde{\mathfrak{t}}') = p_1(\mathfrak{t}') - 1$. The exceptional evaluations are ∓ 1 , while (139) gives exponents $o_{\tilde{\mathfrak{t}}'}(w + \text{PD}[e], \mathfrak{s}^+) = o_{\mathfrak{t}'}(w, \mathfrak{s}) - 1$ and $o_{\tilde{\mathfrak{t}}'}(w + \text{PD}[e], \mathfrak{s}^-) = o_{\mathfrak{t}'}(w, \mathfrak{s})$. The two exceptional contributions therefore add after orientation weighting. Their factor two converts $2^{-(\delta+1)}$ to $2^{-\delta}$.

Algebraically,

$$\begin{aligned} & 3(\beta^2 - 1) + (c_1^2(X) - 1) + 2\beta \cdot c_1(\mathfrak{t}') + 4(\delta + 1 - 2m) - 4m \\ &= 3\beta^2 + c_1^2(X) + 2\beta \cdot c_1(\mathfrak{t}') + 4(\delta - 2m) - 4m. \end{aligned}$$

Together with $Q_{\tilde{X}}(e, h) = 0$ and $e^2 = -1$, this is the $k = 0$ case of [FL02, Proposition 6.5].

Corollary 8.4 (degree consequence). *Under the additional abundance, effectiveness, Seiberg–Witten-simple-type, and characteristic-data assumptions in Theorem C,*

$$\begin{aligned} (158) \quad & \mathcal{D}_X^w(h) \equiv 0 \equiv \text{SW}_X^w(h) \pmod{h^{c(X)-2}}, \\ & \mathcal{D}_X^w(h) \equiv 2^{2-c(X)} e^{Q_X(h, h)/2} \text{SW}_X^w(h) \pmod{h^{c(X)+2}}. \end{aligned}$$

In the last two relevant degrees,

$$\begin{aligned} (159) \quad & D_X^w(h^{c(X)-2}x) = 2^{3-c(X)} \sum_{\mathfrak{s}} (-1)^{(w^2 + w \cdot c_1(\mathfrak{s}))/2} \text{SW}_X(\mathfrak{s}) \langle c_1(\mathfrak{s}), h \rangle^{c(X)-2}, \\ & D_X^w(h^{c(X)}) = 2^{2-c(X)} \sum_{\mathfrak{s}} (-1)^{(w^2 + w \cdot c_1(\mathfrak{s}))/2} \text{SW}_X(\mathfrak{s}) \\ & \quad \times \left(\langle c_1(\mathfrak{s}), h \rangle^{c(X)} + \binom{c(X)}{2} \langle c_1(\mathfrak{s}), h \rangle^{c(X)-2} Q_X(h, h) \right). \end{aligned}$$

Proof. For abundant, effective manifolds of Seiberg–Witten simple type, choose the class Λ and lift w as in [FL02, Theorem 1.1]. Then $r(\Lambda) = c(X) - 4$ and $i(\Lambda) = c(X) + 4$. The published top-level pairing [FL01b], Theorem 8.2, the Donaldson and Seiberg–Witten blow-up formulas, and the superconformal-simple-type vanishing used in [FL02, Section 6] give (158). Extracting the coefficients of the two final degrees gives (159). $\square$

As a normalization test, for a K3 surface one has $c(X) = 2$ and only the zero basic class, with Seiberg–Witten invariant one in the standard homology orientation. Formula (159) becomes

$$(160) \quad D_{K3}(x) = 2, \quad D_{K3}(h^2) = Q_{K3}(h, h),$$

the constant and quadratic coefficients of $e^{Q_{K3}(h, h)/2}$ in the convention with insertion $1 + x/2$. Thus the global test detects neither an additional sign nor an additional factor two.

9. SCOPE AND FURTHER PROBLEMS

Theorem 2.1 supplies the analytic and orientation input that was deferred in the one-bubble calculation of [FL02]: the corrected chart is continuous and embedded, its zero set is an actual Uhlenbeck neighborhood, and its determinant and boundary orientations agree with those used in the pairing. The proof also shows why the compactified instanton obstruction must be treated as a cone pseudo-bundle and why transversality stops at the middle stratum.

The compact background is permitted to have arbitrary dimension, but its regularity and nonzero-spinor hypotheses are essential here. The proof does not treat a zero-spinor pair quotient or a nonregular base. It avoids the varying-rank cutoff problem by global finite-dimensional stabilization; spectral bundles are used only on positive-scale charts where their rank is locally constant.

There is exactly one charge-one bubble. Consequently there are no collision diagonals, no competing partitions of a symmetric product, and no compatibility maps between gluing charts. The topological overlap machinery of [FL18] addresses those phenomena, but its analytic neighborhood identification remains a separate issue. The present theorem does not imply the corresponding result at level two or higher.

Corollary 8.4 has an equally precise logical range. It removes the deferred neighborhood theorem from the level-one contributions. It continues to use the published top-level pairing, the Donaldson and Seiberg–Witten blow-up formulas, the superconformal-simple-type vanishing input, and the assumptions of abundance, effectiveness, and Seiberg–Witten simple type. In particular, it is neither a proof of the full Witten conjecture nor a removal of the effectiveness hypothesis.

REFERENCES

- [Don87] S. K. Donaldson, *The orientation of Yang–Mills moduli spaces and 4-manifold topology*, J. Differential Geom. **26** (1987), no. 3, 397–428.
- [DK90] S. K. Donaldson and P. B. Kronheimer, *The geometry of four-manifolds*, Oxford Mathematical Monographs, Clarendon Press, Oxford, 1990.
- [Fee01] P. M. N. Feehan, *Critical-exponent Sobolev norms and the slice theorem for the quotient space of connections*, Pacific J. Math. **200** (2001), no. 1, 71–118; arXiv:dg-ga/9711004.
- [FL98] P. M. N. Feehan and T. G. Leness, *$PU(2)$ monopoles. I: Regularity, Uhlenbeck compactness, and transversality*, J. Differential Geom. **49** (1998), no. 2, 265–410; arXiv:dg-ga/9710032.
- [FL01a] P. M. N. Feehan and T. G. Leness, *$PU(2)$ monopoles and links of top-level Seiberg–Witten moduli spaces*, J. Reine Angew. Math. **538** (2001), 57–133; arXiv:math/0007190.
- [FL01b] P. M. N. Feehan and T. G. Leness, *$PU(2)$ monopoles. II: Top-level Seiberg–Witten moduli spaces and Witten’s conjecture in low degrees*, J. Reine Angew. Math. **538** (2001), 135–212; arXiv:dg-ga/9712005.
- [FL99] P. M. N. Feehan and T. G. Leness, *$PU(2)$ monopoles. III: Existence of gluing and obstruction maps*, preprint, 1999; arXiv:math/9907107.
- [FL02] P. M. N. Feehan and T. G. Leness, *$SO(3)$ monopoles, level-one Seiberg–Witten moduli spaces, and Witten’s conjecture in low degrees*, Topology Appl. **124** (2002), no. 2, 221–326; arXiv:math/0106238.
- [FL18] P. M. N. Feehan and T. G. Leness, *An $SO(3)$ -monopole cobordism formula relating Donaldson and Seiberg–Witten invariants*, Mem. Amer. Math. Soc. **256** (2018), no. 1226; arXiv:math/0203047.
- [Uhl82] K. K. Uhlenbeck, *Connections with L^p bounds on curvature*, Comm. Math. Phys. **83** (1982), no. 1, 31–42.

BERND JOHANNES WUEBBEN, NEW YORK, NY, wuebben@gmail.com